Romanized Hindi Is Harder Than Hinglish: A Scoring-Aware Audit of Small LLMs on Mathematical Word Problems

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Certificate

I hereby certify that the project report entitled **"Romanized Hindi Is Harder Than Hinglish: A Scoring-Aware Audit of Small LLMs on Mathematical Word Problems"**, submitted by Divyansh Raj (23/MC/195), Ayush Kumar Singh (23/MC/194) and Divyanshu Mishra (23/MC/196) to the Department of Applied Mathematics, Delhi Technological University, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a record of the project work carried out by the students under my supervision. To the best of my knowledge, this work has not been submitted in part or in full for any degree or diploma at this university or elsewhere.

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Candidate's Declaration

We, Divyansh Raj (23/MC/195), Ayush Kumar Singh (23/MC/194) and Divyanshu Mishra (23/MC/196), students of B.Tech. (Mathematics and Computing), hereby declare that the project report entitled **"Romanized Hindi Is Harder Than Hinglish: A Scoring-Aware Audit of Small LLMs on Mathematical Word Problems"**, submitted to the Department of Applied Mathematics, Delhi Technological University, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is original and has not been submitted, in part or in full, for any other degree or diploma.

The use of AI assistants in the project is disclosed in full in Section 10.3. Human reviewers made every dataset acceptance decision, and we take responsibility for the content of this report.

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[*Add any other acknowledgements (family, friends, compute providers).*]

Abstract

Many Indian users ask mathematical questions in Hinglish, a mix of Hindi and English usually written in Roman script. This project asks whether the language form of an otherwise identical word problem changes whether a small open large language model (LLM) solves it. Each problem is rendered in three matched forms: English (**E**), Romanized Hindi (**R**) and Roman-script Hinglish (**C**). Gold answers are computed in code, the renderings of the newly drafted sets are solved blind by two LLMs, and bilingual humans review the families. Models are evaluated unchanged (4-bit NF4 on a Colab T4) with one fixed English instruction, greedy decoding and a frozen strict scorer, under analysis plans written before any output on each cohort.

On 36 permanently excluded development fixtures, a plausible answer-format instruction made 10/36 Qwen3-4B outputs fail strict scoring and pushed 8/12 R explanations into Devanagari. After one disclosed prompt revision both counts were zero, and a seven-model screen showed that format compliance is strongly model-specific; three models passed a pre-declared inclusion rule.

In two separately reviewed, pre-registered cohorts, which are never pooled, R is the lowest-scoring form for all three backbones. On the primary 70-family Pilot-B cohort, Qwen3-4B scores 94.3/85.7/94.3% on E/R/C, Qwen3-8B 98.6/91.4/98.6% and Llama-3.2-3B 94.3/57.1/70.0%. A 24-family cohort from the original pilot shows the same direction, with Qwen near ceiling. Both Qwen models have equal observed aggregate accuracy on E and C in both cohorts, which is not an equivalence result. The R errors are not concentrated in items that contain Hindi operation words. An audit of the primary model's outputs places most substantive errors in R and shows English-explanation compliance falling from 70/70 (E) to 53/70 (R).

The ordering R $<$ C is not new: prior work reports it on translated GSM8K items. This project contributes newly drafted, human-reviewed triplets, a scoring-aware frozen protocol, two separately reviewed cohorts and a reproducible pipeline. It also prepares 300 frozen families for a fine-tuning study.

In that pre-registered follow-up, Qwen3-4B was QLoRA fine-tuned on a free Colab T4 in three arms (E+E, E+R, E+C) with matched content, matched exposure and identical English targets, with three paired seeds, and was evaluated once on 100 untouched test families. Hinglish training did not beat Romanized-Hindi training on Hinglish questions: $\Delta_C = -4.33$ points, 95% interval $[-8.00, -0.67]$, which the pre-registered rule labels *Harm*; all three seeds are negative. Every trained arm also lost about 24–26 points of English accuracy against the unchanged model, consistent with its short training targets. A human-checked audit shows that the extra E+C failures are substantive errors, not formatting failures.

Keywords: Hinglish, Romanized Hindi, code-mixing, mathematical reasoning, LLM evaluation, answer extraction, pre-registration, QLoRA fine-tuning.

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List of Abbreviations

Term	Meaning
A01	Pseudonym of the adjudicator of the original pilot review
ARR	ACL Rolling Review
BBPE	Byte-level byte-pair encoding
C	Roman-script Hinglish (code-mixed) rendering
CC BY 4.0	Creative Commons Attribution 4.0 licence
CI	Confidence interval
CMI	Code-Mixing Index
CSV	Comma-separated values
D001–D043	Entries in the project decision log (`docs/DECISIONS.md`)
DEV	Phase-2 development split
E	English rendering
EACL	Conference of the European Chapter of the ACL
FINAL_TEST	Phase-2 held-out test split, evaluated once, after the adapter freeze
FMT2	The 12 excluded format-test fixture families (36 prompts)
FP16	16-bit floating point
GPU	Graphics processing unit
GSM8K	Grade School Math 8K benchmark
JSON / JSONL	JavaScript Object Notation / one JSON object per line
LLM	Large language model
LoRA / QLoRA	Low-rank adaptation / quantized LoRA
MGSM	Multilingual Grade School Math
MDE	Minimum detectable effect
MIT	The MIT software licence
NF4	4-bit NormalFloat quantization
pp	Percentage points
R	Romanized Hindi rendering
R01, R02	Pseudonyms of the two independent solvers of the original pilot (R02 is also its human language reviewer; R01's AI-prefilled stage-2 judgments are excluded)
RQ	Research question
SDPA	Scaled dot-product attention
SHA-256	Secure Hash Algorithm, 256-bit digest

Term	Meaning
SRW	Student Research Workshop
T4	NVIDIA Tesla T4 GPU (Google Colab)
TRAIN	Phase-2 training split

Chapter 1

Introduction

1.1 Motivation

A model that answers "How many marbles does Aman have?" may fail on *Aman ke paas kitne kanche hain?* although the mathematics is identical. Hundreds of millions of people write Hindi in the Latin alphabet in messages, search queries and chat, often mixing in English words. If an assistant becomes less reliable when a student asks the same question in this everyday form, the reliability gap falls on exactly the users that such assistants are meant to reach.

Measuring this gap is harder than it looks. If accuracy drops on a Hinglish question, the cause could be Hindi vocabulary, Roman-script spelling, the mixing of languages within a sentence, a harder or ambiguous wording, or the evaluation protocol itself: a correct answer written in an unexpected format can be scored as wrong. Comparing unrelated English and Hinglish questions cannot separate these causes. This project therefore holds the mathematics fixed and varies only the language form.

1.2 The E/R/C problem

Each mathematical *family* is one problem rendered three times:

- **E**: English;
- **R**: Hindi phrasing written in Roman script, with a short borrowing list for numbers, units and established loanwords;
- **C**: Roman-script Hindi with English lexical material beyond that list (Hinglish).

A model sees each rendering in a fresh context, with one fixed English instruction, and is scored by one frozen scorer. Because the three renderings share their mathematics and gold answer, differences in correctness across E, R and C are differences in how the model handles the language form, conditional on the renderings being equivalent and natural. Ensuring that condition is the job of the review process described in Chapter 3.

1.3 Research questions

The project has two phases. Phase 1 (Chapters 3–7) answers the first question; Phase 2, the training study (Chapters 8 and 9), answers the other two.

RQ1 (Phase 1)

How does an unchanged model's correctness differ across mathematically equivalent E, R and C prompts?

RQ2 (training study)

Does E+C fine-tuning improve held-out C accuracy over matched E+R fine-tuning?

RQ3 (training study)

Do any gains transfer to unseen phrasing and spellings, and what happens to English accuracy?

1.4 Contributions

The ordering found here, with Romanized Hindi below Hinglish, is not a new discovery. CodeMixTax [1] aligns the same 955 GSM8K questions across five forms and already reports Romanized Hindi below Romanized Hinglish; this is direct prior evidence for the main result. Peter et al. [2] show that translation errors and answer extraction can distort multilingual math gaps. Building on that evidence, this project provides a reproducible, scoring-aware audit of small open LLMs. Its contributions are:

1. **New triplets.** The 370 frozen families were drafted rather than derived from GSM8K and are written in natural Roman spelling rather than scheme transliteration. Every family has code-computed gold and blind dual-model verification, and all evaluated cohorts were human-reviewed.
2. **A scoring-aware frozen protocol.** On excluded fixtures, a plausible final-answer instruction caused form-dependent scoring failures and language drift. One disclosed revision removed both, and a seven-model compliance screen with a predeclared rule preceded model admission.
3. **Two separately reviewed, pre-registered cohorts**, analysed and reported separately, whose direction is consistent: R is the lowest-scoring form for all three backbones.
4. **A negative result on a plausible mechanism.** The development data suggested that Hindi operation words (*bhaag*, *bhinn*) cause R errors. A balanced keep/switch design tested this, and the R errors were not concentrated in such items.
5. **A reproducible system**: self-contained Colab notebooks with an append-only journal and resume, a frozen scorer plus a pre-registered secondary endpoint, checksummed result exports, an independent table builder, a blind-solver toolkit and an authoring kit, with 161 passing software tests.
6. **A controlled, pre-registered fine-tuning comparison.** With matched content, exposure and identical English targets, Hinglish (E+C) training did not improve Hinglish accuracy over Romanized-Hindi (E+R) training under the frozen recipe.

The registered label is Harm ($\Delta_C = -4.33$ points $[-8.00, -0.67]$), confirmed by an independent re-computation and a human-checked failure audit.

1.5 Report outline

Chapter 2 reviews related work. Chapter 3 defines the forms, describes how the datasets were drafted, verified and reviewed, and lists the frozen datasets. Chapter 4 describes the evaluation software and its testing. Chapter 5 reports the development findings on excluded fixtures that shaped the protocol. Chapter 6 reports the results for both cohorts and the output audit, followed by two clearly labelled post-hoc exploratory analyses. Chapter 7 discusses them. Chapter 8 records the Phase-2 design and feasibility work, and Chapter 9 reports the executed fine-tuning study. Chapter 10 sets out limitations and ethics, and Chapter 11 concludes. Appendices give the frozen prompt, dataset hashes, the decision log, the review forms and reproduction instructions.

Throughout, every result carries an *evidence label*: software verification, development evidence on excluded fixtures, or a cohort result. Cohorts are never pooled.

Chapter 2

Literature Review

This chapter places the project among prior work on multilingual mathematical reasoning, code-mixed and Romanized evaluation, and the measurement problems that affect both. The citations come from a bibliography whose entries were each checked against a primary source (ACL Anthology, arXiv, DOI records or publisher pages). A bounded novelty refresh on 26 October 2026 inspected fifteen recent works; it is a targeted primary-source audit, not an exhaustive review.

2.1 Mathematical word problems and multilingual reasoning

GSM8K [3] is the standard English grade-school benchmark: 7,473 training and 1,319 test problems, each with a numeric answer. MGSM [4] translates 250 GSM8K test problems into ten languages by hand and shows that chain-of-thought reasoning transfers across languages, but it contains no Hindi.

For Hindi, GSM8K-Hi [5] is a machine translation of the GSM8K test set into Devanagari, reviewed and corrected by humans. HAWP [6] contains 2,336 Hindi arithmetic word problems with equation annotations; its authors document wrong word senses and translation errors during construction. Anand et al. [7] fine-tune open models on Hindi and English mathematics, explicitly excluding Romanized Hindi. Several studies train for multilingual mathematical reasoning: mCoT [8] targets reasoning consistency across languages, MathOctopus [9] studies multilingual instruction data, and XCOT [10] uses cross-lingual instruction tuning while keeping English responses for translated queries. None of these isolates the Roman-script forms that Indian users commonly type.

2.2 Code-mixed and Romanized evaluation

CodeMixTax. The closest prior work is the CodeMixTax report [1]. It aligns the same 955 GSM8K questions (and MMLU items) across five forms (English, Devanagari Hindi, mixed-script Hinglish, Romanized Hinglish and Romanized Hindi) and evaluates 16 models. Romanized Hindi falls below Romanized Hinglish; for example, it reports E/Romanized-Hinglish/Romanized-Hindi maths accuracy of 90.26/66.18/40.63 for Gemma-3-4B. Its Romanized forms come from deterministic scheme-style romanization of machine-translated Hindi, and the report acknowledges semantic translation drift. It

is an author research report, not a peer-reviewed paper. This project builds directly on it: the ordering is the same, while the items, spelling style, review and scoring protocol differ.

CodeMixBench and Indi-RomCoM. CodeMixBench [11] evaluates code-mixing across 18 languages and eight tasks. Its Hindi–English GSM8K variant mixes Devanagari Hindi with Latin-script English, so it is not a Roman-script comparison, and tested models score lower on it than on English. Indi-RomCoM [12] (a preprint) varies the intensity of Romanized code-mixing (25/50/75%) for four Indic languages on tasks including GSM8K, but has no separate monolingual Romanized-language condition, and asks for output in the input language and script.

Romanized input elsewhere. Sarvam-M [13], an author technical report, shows GSM8K accuracy falling from native-script to Romanized Indic input. Romanized input also hurts LLM triage of real health messages in Indian languages [14]. These results support the existence of a script or romanization penalty but do not include a matched code-mixed condition.

2.3 Measurement: translation errors and answer extraction

Peter et al. [2] audit MGSM and show that translation errors, ambiguous source questions and answer-extraction details (for example, numerals in another script) can create or remove apparent language gaps; after repairs, many gaps shrink. This is the methodological predecessor of the present project's scoring-aware design. It does not license changing a frozen scorer after seeing results; it motivates checking the scorer on excluded data *before* the evaluation, which is what Chapter 5 reports.

2.4 Operation cues, spelling and tokenization

Mathematical solvers can exploit shallow cues: SVAMP [15] showed that English word-problem solvers performed well even without the question text, and GSM-Symbolic [16] shows that LLMs turn irrelevant phrases into operations. This motivates the operation-word hypothesis tested here: that Hindi words such as *bhinn* ("fraction", but also "different") could be misread.

Romanized Hindi has no standard orthography. Code-mixed social-media text shows systematic borrowing and mixing [17], and the Dakshina dataset collects multiple attested

Table 2.1: Closest prior work and how this project differs (from the verified related-work audit).

Work	What it measures	Difference from this project
CodeMixTax	955 GSM8K items in five aligned forms; 16 models; R below Romanized Hinglish	Translated GSM8K with scheme romanization; this project uses new drafted items, natural spelling, human review and a frozen strict scorer
CodeMixBench	Hindi–English GSM8K among 18 languages	Mixed Devanagari/Latin script; no monolingual R arm
Indi-RomCoM	Romanized code-mixing at 25/50/75%	No separate monolingual R condition; output in input language
Peter et al.	Translation and extraction errors in MGSM	Not an R/C study; methodological predecessor of the scoring audit
Sarvam-M	Native vs Romanized Indic GSM8K	No matched code-mixed arm

romanizations per word and explains why round-trip reconstruction of the native script is imperfect [18]. Each R rendering in this project is therefore one of many possible spellings.

Tokenizers split some languages into many more tokens than others, with parallel text differing by up to 15 times in length [19]. The Qwen3 models [20] use byte-level BPE with a 151,669-token vocabulary. This project treats token counts as correlates to report, not as causes.

2.5 Efficient inference and fine-tuning

All evaluation uses 4-bit NF4 quantization with double quantization, introduced with QLoRA [21], which allows 4–8B models to run on a free 16 GB T4 GPU. The Phase-2 training study uses QLoRA adapters on the same quantized stack.

2.6 Positioning

Table 2.1 summarizes how the closest works differ from this project. The broad "Romanized Hindi is harder than Hinglish" ordering is already present in prior work, so this project is positioned as a focused replication and extension with an evaluation audit, not as the first observation of that ordering. The specific combination examined here was not found in the inspected sources: newly drafted human-reviewed triplets, small Qwen and Llama backbones, a frozen strict-plus-numeric scoring protocol and two separately reviewed cohorts. That absence is a retrieval limit, not a guarantee of novelty. The contribution of the training phase is a controlled E+E/E+R/E+C fine-tuning comparison with matched content and identical English targets. A bounded audit of 14 works on 10 October 2026 did not locate this exact comparison.

Chapter 3

Methodology and Dataset

3.1 Language forms

E (English). One unambiguous English question with a single numeric answer and, where relevant, a clear unit.

R (Romanized Hindi). Hindi phrasing in Roman script. A *borrowing list* allows digits, symbols, listed units, INR notation, names and established loanwords. Decision D033 extended this list with *type karna*, *cousin* and loan nouns the reviewer accepted in R (for example *samosa*, *petrol*, *auto*, *taxi*, *printer*, *computer*). R contains no substantive English clauses or relations.

C (Roman-script Hinglish), two definitions. The original pilot used a *clause-based* definition: C must contain a productive English predicate, clause or mathematical relation alongside Hindi, and English object nouns alone do not qualify. Human review of Pilot-B then showed that natural Hinglish is mostly *insertional*: Hindi grammar with English nouns, conjunct verbs and relation words, as in *apples aur bananas ka ratio 3:7 hai* or *23 add karta hai*. Full English clauses inside Hindi sentences read as unnatural. Decision D033 therefore redefined C for Pilot-B and Phase 2: C must contain English lexical material (nouns, conjunct verbs, relation words) beyond R's borrowing list, but no full English clause is required. The original cohort keeps the clause-based definition. Because the two cohorts use different C definitions, they are always reported separately.

3.2 Family design

A family is one problem with three renderings, so every analysis is paired at the family level: 210 prompts from 70 families are 70 independent units, not 210. Six topics are covered: comparison and remaining quantities, two-step money, fractions and percentages, ratios, rates and units, and elementary algebra. The original pilot has five families per topic; Pilot-B has 12/12/12/12/11/11.

Every family stores its gold value as an exact rational string (e.g. `"1/3"`), its canonical unit, unit aliases and conversions, whether the unit may be omitted, a reference equation and an English reference solution. The authoring checker (Section 4.4) requires that every

arithmetic segment of the final statement of the reference equation equal the gold value exactly, using rational arithmetic.

3.3 AI drafting with code-computed gold

Decision D027 scaled the dataset beyond the original pilot by drafting families with AI and computing gold in code.

- **Original pilot (30 families).** Per the author's provenance statement (D038), the English statements were written by the project author and the R and C renderings were produced by AI translation (OpenAI GPT via Codex), then reviewed by humans. The draft metadata of 10 October records `AI_DRAFT` for all 30 families; per the author that label is mistaken for the E texts. The frozen files were left unchanged so their hashes stay valid, and the correction is recorded in `pilot_original/FROZEN.md`.
- **Pilot-B (70).** Drafted by Claude (`claude-opus-5-5`).
- **Phase-2 FINAL_TEST (100) and DEV (40).** Drafted by GPT (`gpt-6-astra`).
- **Phase-2 TRAIN (160).** Drafted by Claude.

The split of drafters (GPT for test and development data, Claude for training data) was a safeguard against a training set and test set sharing one drafter's style.

3.4 Blind dual-model verification

Before human review, every rendering of the AI-drafted sets was solved blind by two LLMs, Claude and GPT, each in a fresh context with only the rendering text (Section 4.4). No Qwen model ever drafted, solved or checked any item, because Qwen3-4B is the primary model under evaluation. Each solver returns an exact final answer, unit, the quantities it extracted, the entities and the asked-for quantity, plus an optional ambiguity note. The comparison flags gold mismatches, solver disagreements, cross-form differences in extracted quantities or asked-for target, ambiguity notes and missing outputs.

- **Pilot-B:** both solvers matched gold on 210/210 first-version renderings and on 15/15 revised renderings.
- **Phase 2 (300 families, 900 renderings):** Claude 900/900 and GPT 897/900; all three misses were in one family (TRA072) and were a GPT error. Every rendering changed in FINAL_TEST v3 was re-verified by both solvers (204/204 each), and every rendering changed in TRAIN/DEV v2 likewise (43/43 each).

Solver agreement shows only that the text encodes the intended calculation. It does not certify naturalness or independence, and the two solvers may share blind spots.

3.5 Human review

3.5.1 Pilot-B: single reviewer

One bilingual team member reviewed all 70 Pilot-B families (equivalence, naturalness of R and C, and a decision), with gold hidden until after their own assessment. The outcome was 57 ACCEPT, 13 REVISE and 0 REJECT, with two equivalence failures. The reviewer's fixes were applied verbatim as version 2 renderings; version 1 is kept unchanged. One consequential edit by Claude (PB065-C: *chachere bhai* $\rightarrow$ *cousin*) was recorded as pending reviewer confirmation. Family PB051 (a 10:05–10:25 time interval) was accepted despite a solver ambiguity note; its gold assumes 20 minutes. Pilot-B is therefore *single-reviewer* data, and no second annotator has yet reviewed it.

3.5.2 Phase-2 FINAL_TEST, TRAIN and DEV

The same reviewer reviewed all 100 FINAL_TEST families: 32 ACCEPT and 68 REVISE. Fixes were applied verbatim (v2). Claude then made consequential edits and replaced English verbs in R with everyday Hindi verbs to keep R distinct from C; the reviewer checked these (53 OK, 30 EDIT, 5 MINE), giving v3. In one later case (FT057-C) Claude restored a distractor quantity omitted in C; its gold of 30 was checked by hand but not blind re-solved. Of the 68 revisions, 62 targeted R.

The main review finding was that GPT-drafted Romanized Hindi was systematically over-formal: 68/100 FINAL_TEST families needed repair, against 13/70 for Claude-drafted Pilot-B. This comparison is descriptive, because the drafter is confounded with split, drafting prompt and review timing.

TRAIN and DEV received *sample-based* review: the 6 strictly flagged families plus a seeded 20% random sample of the rest, 45 families in all, giving 19 ACCEPT and 26 REVISE. Repair needs were heavy in GPT-drafted DEV (10 of 11 sampled families revised) and moderate in Claude-drafted TRAIN (16 of 34). The unsampled families are accepted on automated verification plus the sample, so they may still contain stilted Romanized Hindi.

3.5.3 Original cohort: independent solvers, one language reviewer and an adjudicator

The original 30 families went through a two-stage protocol designed for two independent reviewers (R01, R02) and an adjudicator (A01). In stage 1 each reviewer solved all 90 renderings in randomized order *before* seeing any reference answer. In stage 2 the reviewer imported their stage-1 file and rated equivalence, ambiguity, classification and naturalness of each triplet.

The final evidence for this cohort, as recorded in decisions D036 and D037, is:

- **Mathematics: two independent human solvers.** Across all 90 renderings the two reviewers' answers agreed on 80. On the cohort's 72 renderings they agreed on 65, and R02 matched gold on all 72. The six R01 answers that differed from gold were resolved by A01, who confirmed the gold of every cohort family.
- **Language and equivalence: one human plus the adjudicator.** R01's official stage-2 file marks 70 of 90 rows (52 of the cohort's 72) as unverified AI-prefill suggestions, a field that the review form never writes. These judgments are recorded as `ai_prefill_unverified` and excluded from every human-review claim and agreement statistic. Language review of the cohort therefore rests on R02 and A01, with R01's 20 human stage-2 judgments in the cohort kept.
- **Adjudication.** A01 decided 24 ACCEPT, 5 REVISE and 1 EXCLUDE (PIL010, a near-duplicate of PIL002).
- **Cohort.** The two returned re-reviews of the five repaired families were byte-identical, so they could not count as independent re-review. The cohort is the 24 families accepted *without repair*: PIL001–PIL008, PIL011–PIL014, PIL016–PIL018, PIL021–PIL023 and PIL025–PIL030.
- **Discarded rounds.** Earlier review rounds were found not to be independent and were discarded; they are archived as unused, together with later edited copies.

The conversion of these review files into the frozen cohort was cross-checked against an independent implementation written by a second AI assistant (GPT, "Sol"): cohort, text hashes, answers and decisions all matched.

3.6 The operation-word lexicon

The two R errors in the development study (Chapter 5) involved *bhaag* ("divide") and *bhinn* ("fraction"). An AI research assistant therefore proposed a bounded lexicon of 25 Hindi operation-word families with spelling variants. Each entry is flagged either *A* (lexically ambiguous: polysemy or Roman spelling collision) or *C* (context-dependent relation). Examples:

- *A*: *bhinn* (fraction / different), *bhaag* (divide, part / run), *guna* (multiply / quality), *bacha* (remained / child under consonant simplification), *kul* (total / family), *varg* (square / class);
- *C*: *kam* (less), *zyada* (more, and an upper bound in *zyada se zyada*), *anupat* (ratio), *pratishat* (percent), *prati* (per).

One bilingual team member reviewed all entries; 13 were AI- or dictionary-assisted and then endorsed by the reviewer, and *anupat* (ratio) remains unconfirmed. The ambiguity of *bhinn* exists even in Devanagari: a Hindi–English dictionary lists both senses.

A fixed matcher (`operation_vocab_draft_v1`: NFC case-folded text, maximal letter

Table 3.1: Frozen datasets. All were frozen on 16 October 2026 except the original cohort (D036, 26 October 2026).

Set	Families	Drafted by	Human review	Status / use
Original cohort	24 of 30	E: author; R/C: GPT translation	Two independent solvers; language review R02 + A01	Evaluated (Ch. 6)
Pilot-B	70	Claude	One reviewer, all families (57 ACC, 13 REV)	Evaluated; primary cohort
FINAL_TEST	100	GPT	One reviewer, all families (32 ACC, 68 REV)	Untouched until confirmatory evaluation
DEV	40	GPT	Sample-based (shared 45-family sample)	Training study
TRAIN	160	Claude	Sample-based (shared 45-family sample)	Training study (train_A 80, train_B 80)
Pilot-B + Phase 2	370			

sequences, listed surfaces only, longest phrase first, no stemming) was frozen before any pilot output. On the original 30 drafts it finds a lexicon word in 27/30 R but only 10/30 C renderings (A-family words: 19/30 R, 5/30 C; *bhaag*/*bhinn*: 2/30 R, 0/30 C; E: 0/30). On the original set, a C advantage could therefore partly reflect easier vocabulary.

Keep/switch balance in Pilot-B. Pilot-B was drafted to counter this. In 18 families C *keeps* R's Hindi operation word, in 17 C *switches* it to English, and 35 families are controls with no Hindi operation word in R. These tags were taken from the frozen family provenance and fixed before any Pilot-B output.

3.7 Frozen datasets

Table 3.1 lists every frozen dataset. Pilot-B and Phase 2 together give **370 frozen families**; the 24-family original cohort is separate. Each file has a recorded SHA-256 hash (Appendix B), and any text change requires a new version and a new hash. The 12 format fixtures (FMT2) and all 30 original pilot families are permanently excluded from training and from the final test.

3.8 Evaluation protocol

Prompt. One English system instruction, `test1-english-final-v2`, is used for every form and model (Appendix A). It asks for a short explanation in English and a final line beginning `FINAL:`, followed by an English unit only when the answer has one and by

nothing after a pure number. The user message is the question text alone. The instruction was revised once, on excluded data, before any cohort output (Chapter 5).

Models and decoding. Qwen3-4B [20] is the primary backbone; Qwen3-8B and Llama-3.2-3B-Instruct are secondary, descriptive backbones under the identical protocol. Every model is loaded at a pinned immutable revision with NF4 double quantization, FP16 compute, the SDPA math kernel, batch size 1, greedy non-thinking decoding (thinking disabled for Qwen), a 1,024-token output cap, a fresh context per prompt and a seeded order. One infrastructure retry is allowed; answer-dependent retries are not. Llama's chat-template date is pinned to 15 October 2026 so that its prompts stay frozen.

Primary endpoint. A frozen strict scorer (v1.0.0) requires exactly one terminal `FINAL:` line, an exact rational value and a unit validated against pre-declared aliases and conversions. Missing, refused, ambiguous and truncated answers count as wrong.

Secondary endpoint. `numeric_literal_v1`, adopted before the pilot (D018), checks whether the literal value on the sole FINAL line equals gold, with no unit conversion. It is reported beside the strict endpoint, never instead of it. Disagreement between the endpoints does not prove a format-only error, and agreement does not rule out formatting failures that both endpoints reject.

Statistics. Accuracy per form is reported with 95% Wilson intervals. Paired E−R, E−C and R−C differences use 95% percentile intervals from 10,000 family-bootstrap resamples (seed 101), resampling whole triplets, together with discordant-family counts. The cohorts are diagnostic samples, so no significance claims are made.

Pre-registration. Each cohort's analysis plan was written and committed before any model output on that cohort: `pilot_b/PREREGISTRATION.md` (D035) and `pilot_original/PREREGISTRATION.md` with Amendment 1 (D037). The original cohort's plan was written with Pilot-B results already known, so that cohort is a descriptive replication, not an independent confirmation. The plans fix six endpoints: (1) primary strict accuracy for Qwen3-4B; (2) the secondary numeric endpoint and invalid-reason counts; (3) paired differences; (4) the same tables for the secondary backbones; (5) descriptive operation-word strata; (6) after the run, an English-compliance and error-taxonomy audit, plus tokens and latency per form. Results are reported whatever their direction, and the cohorts are never pooled.

Chapter 4

System Design and Implementation

Evidence label: software verification. This chapter describes what the software does and how it was tested. Tests show that the software behaves as specified; they are not research results.

4.1 Architecture overview

Figure 4.1 shows the pipeline from authoring to tables. It has three goals: the gold and the scoring must be computed, never typed by hand; every run must be resumable on an unreliable free GPU without changing the protocol mid-run; and every reported number must be reproducible from checksummed artefacts.

4.2 Evaluation engine and Colab notebooks

The evaluation engine (`test1_pilot/hinglish_pilot/`) has modules for data validation (`data.py`), the run loop (`engine.py`), the Hugging Face backend (`hf_backend.py`), reporting (`report.py`) and the scorer (`scorer_v1.py`, a byte-for-byte copy of the original `pilot/scripts/scorer.py`). A builder script embeds all of this, together with fixtures and documentation, into a single self-contained notebook, so a team member only uploads one `.ipynb` file to Colab and chooses *Run all*. Separate notebooks were built for each cohort and model (`colab_notebooks/pilot_b_eval/`, `pilot_original_eval/`) and for model screening (`model_screening/`). A Pilot-B notebook takes about 25–35 minutes on a T4.

Modes. The engine has four modes, each with its own validation gate:

- `demo`: a fake backend with synthetic fixtures, for software checks only;
- `reviewed`: 30 families with two reviews and adjudication;
- `pilot_b`: 1–200 families with single-reviewer attestation and split role `PILOT_B_EVAL` (D035);
- `pilot_adjudicated`: the original-cohort mode (D036), with `judgment_source` recorded per reviewer form (D037).

Research modes reject the fake backend, DRAFT status and the wrong split role. Every run's outputs carry a label naming its evidence type (for example "PILOT-B EVALUATION — single-reviewer frozen data").

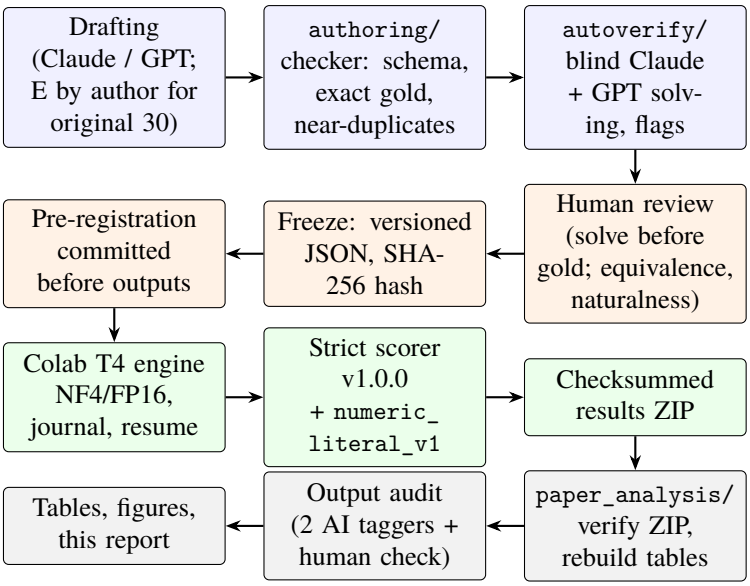


Figure 4.1: Pipeline from drafting to reported tables. Blue: dataset construction; orange: human gate and freeze; green: inference and scoring on Colab; grey: analysis.

Inference recipe. The engine loads the model and tokenizer at immutable revisions resolved on first load, using NF4 with double quantization and FP16 compute on a single CUDA device, with the SDPA math kernel and no adapters. It uses greedy non-thinking decoding with generation seed 11 and deterministic per-item seeds, a 1,024 new-token cap, a total context preflight cap of 2,048 tokens and batch size 1. The verified stack is torch 2.11.0+cu130, transformers 4.51.3 and bitsandbytes 0.50.2. Before scored generation, every prompt is tokenized and checked, so that no question is truncated. Silent changes to the cap or precision, CPU offloading, thinking mode and model substitution are all forbidden.

Journal and resume. Colab sessions disconnect. Each generation attempt is journaled *before* it starts and committed as an atomic JSON file. On resume, the engine reuses frozen input snapshots and immutable revisions. It refuses to continue if the data, configuration, code, backend facts or package environment have changed, and it skips successful generations even when they are wrong, refused, malformed or truncated. At most one infrastructure retry is allowed, with unchanged settings and a preserved record. Out-of-memory errors stop the run instead of being retried. A lock file prevents two writers in one run directory, and a stale lock can be recovered only by an explicit setting. Headline accuracy is suppressed until every required output is complete.

Checksummed export. The results ZIP contains manifests, exact input snapshots and prompts, attempt records, automatic predictions, per-family comparisons, metrics, errors, an audit CSV and a `checksums.sha256` file covering every member. Export uses a fixed allow-list of files and rejects common credential and e-mail patterns. CSV cells that begin with spreadsheet formula characters receive a protective apostrophe (a bug in this guard,

which also prefixed empty strings, was fixed before any pilot run; D022). Manual audit overlays are stored as separate versioned revisions and never overwrite automatic scores.

4.3 Scoring

Strict scorer v1.0.0 (primary). The scorer finds exactly one `FINAL:` line, which must be the last non-blank line, and parses an exact rational value. It then validates the unit text against the family's canonical unit, aliases and conversions. Dimensionless answers must carry no suffix. Missing, duplicate or non-terminal FINAL lines, unsupported values, invalid units, refusals and truncation are all scored wrong, with a recorded `invalid_reason`.

`numeric_literal_v1` (secondary). Implemented in `secondary_analysis/numeric_endpoint.py` and adopted under D018, this endpoint never consults gold during extraction. It requires exactly one FINAL marker at the start of a line (bold variants allowed), never recovers an answer from other lines and fails truncated outputs. Its grammar accepts an optional one-letter assignment (`x =`), an optional currency prefix (₹, INR, Rs., Rs), a scalar optionally wrapped in `**`, `$` or `\boxed{}`, and an optional word suffix. The literal value is compared with gold exactly, with no unit conversion. The grammar has no rule for a currency symbol *after* the number; Chapter 6 shows the consequence.

CMI. The same package implements a Code-Mixing Index (CMI_v1) over token tags that a single bilingual human corrects from heuristic pre-tags. Its output is labelled as heuristic and single-annotator.

4.4 Analysis and verification tools

`paper_analysis/`. This standard-library tool turns results ZIPs into Markdown and LaTeX tables, a combined per-rendering CSV and figures. It adds no new scoring. It treats ZIPs as untrusted: they are read into memory, never extracted or executed, and absolute or `..` paths, symlinks and encrypted, duplicate or oversized members are rejected. Every checksum entry must match, every member must be listed, and attempt record hashes and snapshot hashes are verified against the run manifest. It recomputes accuracy, Wilson intervals and the paired bootstrap with the same formulas and random-number order as the engine, checks them against the stored `metrics.json`, and exits with code 3 on any mismatch. Runs are never pooled. All six cohort runs reproduced the stored metrics exactly.

`autoverify/.` This toolkit makes blind solver tasks and compares the answers. `make_solver_tasks.py` writes one shuffled task file per solver containing only four whitelisted fields (family ID, rendering ID, form, text), so gold, equations and metadata never reach a solver. `compare.py` joins outputs by rendering ID and rejects malformed, duplicate or unknown rows. It flags gold mismatches, solver disagreement, cross-form quantity or target mismatches, ambiguity notes and missing outputs. It then writes a review sheet listing all flagged families plus a seeded random 20% of the unflagged ones, with the gold column placed last so it can be hidden until the reviewer has assessed the question. Automated checks never assign ACCEPT.

`authoring/.` The authoring kit has a family template, a guide and `check_family.py`. The checker validates structure and the gold schema and evaluates the reference equation with a safe AST evaluator in exact fractions. It also finds duplicate texts and flags character 5-gram Jaccard similarity of 0.5 or more against the pilot drafts and the format fixtures, reporting IDs only and never printing pilot gold. The guide's independence rules forbid adapting public benchmarks, forbid derivatives of the pilot and fixtures, treat numeric siblings as the same family, require operation-word balancing for C, and forbid authors from reviewing their own families.

Training preparation (`t4_smoke/`). The training code builds completion-only labels. It masks the rendered generation prompt, including Qwen3's empty think block, and supervises the English solution plus the end token. It also asserts that the target token IDs are identical across E, R and C. A smoke-test script (`t4_smoke/train_smoke.py`) and a self-contained Colab notebook (`colab_notebooks/training_smoke/`) were added on 27 October (D041) and run on a T4 on 28 October (D042; Section 8.1.1). The smoke adapters were discarded and never evaluated; the Phase-2 training runs are described in Chapter 9.

4.5 Testing

All suites were re-run for this report on 26 October 2026 with Python 3.12, without GPU dependencies. Table 4.1 lists the current counts; every test passed. On 15 October the corresponding counts were 22 pipeline, 38 secondary-analysis, 3 scorer and 4 training-protocol tests, and the suites have grown since. With the smoke-test code (D041, 27 October), the `t4_smoke` suite grew to 50 tests, all passing; the table keeps the 26 October counts.

Beyond unit tests, the notebook was validated with nbformat, executed end-to-end in a fresh Jupyter kernel in demo mode, and its exports passed checksum verification. A real-GPU smoke run on a Colab T4 (Qwen3-4B; T4 with 14.56 GiB reported device memory)

Table 4.1: Software test suites (re-run 26 October 2026; all passing).

Suite	What it covers	Tests
`test1_pilot/tests`	Engine: malformed review/split/gold rejection, exact numeric/unit scoring, invalid and truncated outputs, paired bootstrap and discordance, bounded retries, interruption, lost in-flight generation, stale locks, incompatible resumes, journal corruption, overlays, secret detection	23
`secondary_analysis/tests`	`numeric_literal_v1` grammar, CMI, secondary report, notebook freshness	64
`autoverify/tests`	Task whitelisting, seeded shuffles, output validation, flags, review-sheet sampling	23
`t4_smoke`	Training-protocol validation and completion-only labels	21
`paper_analysis/tests`	Safe ZIP reading, checksum verification, table consistency	17
`authoring/tests`	Family checker: schema, equation exactness, near-duplicates	10
`pilot/scripts`	Original exact scorer (including 60 balanced answer cases)	3
Total		161

completed six short generations with no infrastructure errors or truncation. Model load took about 224.75 s (possibly including download), and per-generation reserved memory peaked at about 4.19 GiB. That smoke run's original 0/6 automatic score exposed a fixture error, a blank gold unit on a count question, and the strict suffix rule on fractions. The fixture was repaired in a new demo version and the original scores were preserved. This was the first sign of the format problem studied in Chapter 5.

4.6 Real-browser test of the review forms

The original pilot's reviewers used offline HTML forms (stage 1 solving, stage 2 rating; Appendix D). On 20 October 2026 a script drove headless Google Chrome (via puppeteer-core) through *copies* of the R01 stage-1, R01 stage-2 and R02 stage-2 forms using artificial answers; the distributed packets were not modified. All 14 checks passed with no page JavaScript errors:

- stage 1 loads 90 renderings and shows no gold;
- partial progress downloads and resumes by import;
- completing all solves with timestamps and the attestation exports correctly;
- stage 2 blocks saving before import and rejects an unfinished stage 1;
- stage 2 accepts a finished stage 1, keeps its answers unchanged and records its SHA-256;
- ratings for all 90 rows export;

- R02's stage 2 rejects an R01 file.

The test also produced three usability notes. The stage-1 import message ("Imported answers are read-only in stage2") may confuse reviewers. Progress and final downloads share one filename, so the browser renames repeats and reviewers must return the last file. The packet's C definition is the original clause-based one. The form itself never writes an AI-prefill field, which is how the R01 stage-2 provenance marker was later recognized as external to the form (Section 3.5).

Chapter 5

Development Findings on Excluded Fixtures

Evidence label: development evidence only. All results in this chapter come from 36 AI-authored, unreviewed fixture prompts (12 families × E/R/C, `FMT2`), permanently excluded from both cohorts and from all later training and test data. With 12 prompts per form, every rate is a rough indication, not an estimate.

Before spending the one-shot cohort runs, the project ran the 12 fixture families through the identical pipeline on a Colab T4. The aim was to find protocol problems on data that would never be used as evidence.

5.1 The format confound and the prompt v1→v2 change

The original instruction (`test1-english-final-v1`) asked for `FINAL: <number or fraction> <unit when applicable>`. Under it, 10 of 36 Qwen3-4B outputs were strict-invalid, all with `invalid_unit` (Table 5.1). In 8 outputs, spread roughly equally across forms, the model appended a unit word to a dimensionless answer: `8 unitless`, `24 unit.`, `8 unit (no unit specified, so just the number)` and `7/27 positions.`. In one R output (`109 moti.`) the model echoed the question's Hindi noun, which was not among that fixture's aliases. Since the original pilot has 6 dimensionless families out of 30, the v1 prompt would have failed many correct answers on them in every form. There were no missing, multiple or non-terminal FINAL lines and no refusals.

Decision D020 replaced the instruction with `test1-english-final-v2` (Appendix A): explain in English even when the question is not in English; after FINAL, write a unit in English only if the answer has one, and nothing after a pure number. The v1 text is kept in the code as `INSTRUCTION_V1`, and the scorer was unchanged. Under v2, Qwen3-4B produced 0/36 invalid outputs. Dimensionless answers became bare (`FINAL: 8`) and units became English (`330 rupees`, `59 minutes`). The strict and numeric endpoints agreed on all 36 outputs, and strict-correct counts were 12/10/12.

The fixtures also changed between the two runs. One ambiguous R rendering was repaired (FMT2-001-R, where a fee could be read as per-plate), and the alias *moti* was added; the fixture version went from 0.1.0 to 0.2.0. The 10→0 change is therefore a development comparison, not an isolated prompt effect. The revision was made once, disclosed and versioned. Prompt iteration on the fixtures then stopped, and the prompt was frozen before any cohort generation.

Table 5.1: Qwen3-4B on the 36 excluded FMT2 fixture prompts under prompt v1 and v2 (development evidence; counts of 12 per form).

Prompt	Form	Strict correct	Numeric match	Strict invalid	Devanagari in output	Median output tokens
v1	E	9	12	3	0	82
	R	7	10	4	8	107
	C	9	11	3	0	86
v2	E	12	12	0	0	–
	R	10	10	0	0	–
	C	12	12	0	0	–

5.2 Devanagari drift

Under v1, 8 of 12 R outputs contained Devanagari script despite the instruction to explain in English, and about 10 of 12 contained Romanized-Hindi prose; E had 0/12 and C had about 2/12 with Romanized prose only. Under v2, Devanagari fell to 0/12 R outputs, and a heuristic marker found Romanized-Hindi prose in 1 of 12. The detector is heuristic, not a validated language tag. This drift does not affect either scoring endpoint. It does show that output-language following degrades on Romanized-Hindi input, which made the English-compliance audit of Chapter 6 essential.

5.3 Repeated *bhaag/bhinn* misreadings

Under v2, Qwen3-4B's only two errors were R items. In FMT2-005-R, *bhaag dekar bhinn likho* ("divide, write as a fraction") was answered as a difference. In FMT2-006-R, *kitna bhinn istemaal hua* ("what fraction was used") received the remaining count (20) under both v1 and v2, and Llama-3.2-3B later gave the same 20. Claude, an AI assistant, judged both renderings correct; no independent bilingual check was made. *Bhinn* means both "fraction" and "different", even in Devanagari. With $n = 2$ unreviewed items, these errors are *consistent with* operation-word misreading but do not establish it. They motivated the lexicon and the keep/switch design of Pilot-B (Section 3.6), both fixed before any cohort output and tested in Chapter 6.

5.4 Seven-model format screening and the inclusion rule

Decision D028 screened more unchanged models on the same 36 prompts under the frozen v2 prompt, recipe and scorer. Its pre-declared rule was that a model joins the cohort evaluations only if at most 3/36 fixture outputs are strict-invalid, with no per-model prompt or scorer tuning. Table 5.2 summarizes the screen.

Table 5.2: Format screening on the excluded fixtures (prompt v2; strict-invalid of 36; strict correct of 12 per form). Development evidence. †primary; ×excluded.

Model	Invalid	E/R/C	Dominant failure	Decision
Qwen3-4B†	0	12/10/12	2 R operation-word misreadings	Primary
Qwen3-8B	0	12/12/12	none (1 R output with Devanagari)	Included (D030)
Llama-3.2-3B-Instruct	2	12/9/9	value errors in R and C; 1 truncation, 1 missing FINAL	Included (D032)
Qwen3-1.7B×	36	0/0/0	instruction echo on the FINAL line	Excluded (D029)
Phi-4-mini-instruct×	14	6/4/11	inline FINAL; R repetition to the token cap	Excluded (D025)
Granite-3.3-2B-instruct×	36	0/0/0	commentary after FINAL	Excluded (D026)
Gemma-3-4B-it×	–	–	no outputs (CUDA error under FP16)	Excluded (D031)

Phi-4-mini and Granite were excluded before the $\leq$3/36 rule was formalized; both exceed it. Phi-4-mini's failures show why such a screen is needed. In 7 outputs it wrote the FINAL marker inline at the end of a paragraph, which both endpoints reject, and 6 of its 12 R outputs collapsed into repetition at the 1,024-token cap, in four cases echoing the Hindi question 17–40 times. Its apparent C > E ordering (11/12 vs 6/12) is therefore a formatting artifact. Granite commented on the unit instruction after the answer (e.g. `FINAL: 24, (units: none)`), so most of its numerically right values were rejected. Qwen3-1.7B copied the instruction's wording onto the FINAL line. Gemma-3-4B loaded but every generation failed with a CUDA error under the frozen FP16 recipe; running it would have required a model-specific recipe change, which D028 does not allow.

Two lessons follow. First, within Qwen3, instruction following broke before mathematics did: 36/36 invalid at 1.7B against 0/36 at 4B and 8B. Second, one frozen prompt and scorer cannot simply be applied across model families. Without a compliance check on excluded data, a multi-model comparison would mostly measure formatting. The cohort evaluations therefore use three backbones: Qwen3-4B (primary), Qwen3-8B (within-family scaling) and Llama-3.2-3B (cross-family). The latter two are secondary and descriptive.

Chapter 6

Results

Evidence label: cohort results under pre-registered plans. Pilot-B (70 families, single-reviewer data, insertional C) is primary. The original cohort (24 families, clause-based C) is a descriptive replication whose plan was written with Pilot-B results known. The two cohorts are never pooled. Each backbone ran once per cohort and completed every rendering (210/210 and 72/72), with all checksums verified; the tables reproduce the stored scores exactly.

6.1 Pilot-B: accuracy by form

Table 6.1 gives strict accuracy per form for the three backbones, and Figure 6.1 plots it. R is the lowest-scoring form for every backbone.

6.2 Pilot-B: paired differences

Because the three forms share their families, the relevant comparison is paired (Table 6.2).

1. **R is the weakest form for all three backbones.** The deficit depends strongly on the model: it is modest for Qwen (8.6 and 7.1 points, with intervals crossing or touching zero) and large for Llama (37.1 points, with an interval well above zero).
2. **E and C have equal observed aggregate accuracy for both Qwen models** (E$-$C $= 0.0$). This is not itemwise identity, since the discordances are 3/3 and 1/1, and it is not an equivalence test. C exceeds R, and for Qwen3-4B the R$-$C interval excludes zero (C better in 7 families, R better in 1).
3. **For Llama, C lies between E and R**, 24.3 points below E.
4. **Scale within Qwen.** Going from 4B to 8B raises R from 85.7% to 91.4% and E/C from 94.3% to 98.6%. The E$-$R gap moves only from 8.6 to 7.1 points.

6.3 Pilot-B: operation-word strata

Table 6.3 gives the pre-registered descriptive split by whether C keeps R's Hindi operation word, switches it to English, or R has none, and Figure 6.2 plots the R column. R accuracy is similar across strata for every model, so the R deficit is *not concentrated* in items with *bhaag*/*bhinn*-type vocabulary. The misreading hypothesis suggested by the two development errors is not supported at this scale. Many Llama R errors are arithmetic slips on simple items; for example, in PB017 ($1200 - 3 \times 265 = 405$) Llama answered 435.

Table 6.1: Pilot-B strict accuracy (70 families per form): correct count, accuracy (%) and 95% Wilson interval, and strict-invalid outputs (truncations in parentheses). No refusals occurred.

Backbone	E		R		C	
Qwen3-4B (primary)	66/70	94.3 [86.2, 97.8]	60/70	85.7 [75.7, 92.1]	66/70	94.3 [86.2, 97.8]
invalid		3		2		1
Qwen3-8B	69/70	98.6 [92.3, 99.7]	64/70	91.4 [82.5, 96.0]	69/70	98.6 [92.3, 99.7]
invalid		1		1		0
Llama-3.2-3B	66/70	94.3 [86.2, 97.8]	40/70	57.1 [45.5, 68.1]	49/70	70.0 [58.5, 79.5]
invalid		1		6 (5)		4 (2)

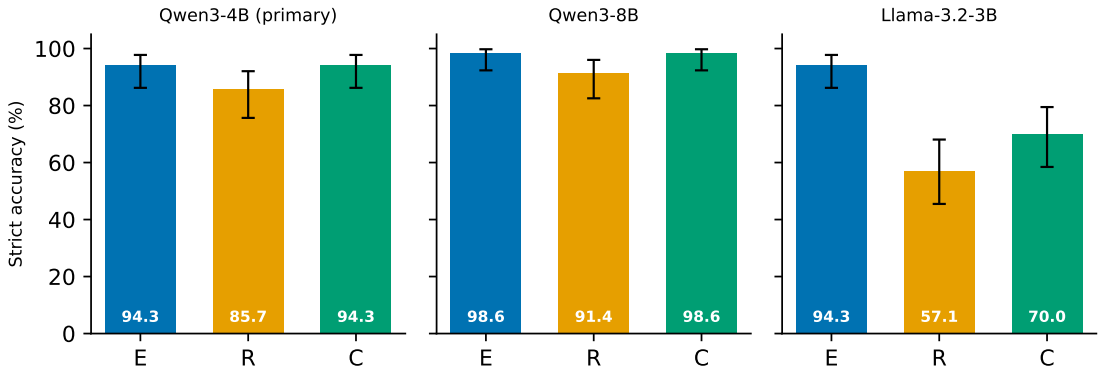


Figure 6.1: Pilot-B strict accuracy by form with 95% Wilson intervals (70 families per bar).

6.4 Pilot-B: residual format effects

Strict-invalid outputs. All six of Qwen3-4B's strict-invalid outputs are `invalid_unit`. Three of its four E failures are numerically correct answers given with English unit phrases outside the aliases (`11 years old`, `13 years old`, `4 bags of cement`). One C failure is an unlisted spelling (`300 rupye`), the form-dependent alias penalty anticipated before the pilot. Together these cost Qwen3-4B four strict points. Qwen3-8B's only strict-invalid E output is `11 years old.` (PB065-E). For Llama, the stored invalid reasons (from `per_rendering.csv`) are: E, one `invalid_numeric_grouping`; R, five truncations and one `invalid_unit`; C, two truncations, one `missing_final` and one `invalid_numeric_grouping`.

Endpoint agreement. The numeric endpoint agrees with strict scoring on all 210 Llama Pilot-B renderings. This does not rule out formatting failures that both endpoints reject: Llama had 11 strict-invalid outputs, 7 of them truncated.

The trailing-₹ gap in the secondary endpoint. For Qwen, the frozen numeric grammar does not accept a *trailing* rupee symbol (`FINAL: 1575 ₹`), which the strict scorer accepts as a unit. This accounts for every strict-correct/numeric-fail case: 30 for Qwen3-4B and 36 for Qwen3-8B. As a result, `numeric_literal_v1` falls *below* strict for both Qwen

Table 6.2: Pilot-B paired family differences (percentage points) with 95% intervals from 10,000 family-bootstrap resamples (seed 101). In parentheses: discordant families correct only in the first / only in the second form.

Backbone	E−R	E−C	R−C
Qwen3-4B	+8.6 [−1.4, +18.6] (9/3)	0.0 [−7.1, +7.1] (3/3)	−8.6 [−15.7, −1.4] (1/7)
Qwen3-8B	+7.1 [0.0, +14.3] (6/1)	0.0 [−4.3, +4.3] (1/1)	−7.1 [−14.3, 0.0] (1/6)
Llama-3.2-3B	+37.1 [+24.3, +50.0] (28/2)	+24.3 [+11.4, +37.1] (21/4)	−12.9 [−27.1, +1.4] (10/19)

Table 6.3: Pilot-B strict accuracy by pre-registered operation-word stratum (descriptive; counts and percentages only).

Backbone	Stratum	Families	E	R	C
Qwen3-4B	C keeps	18	18 (100%)	16 (89%)	17 (94%)
	C switches	17	17 (100%)	14 (82%)	17 (100%)
	none	35	31 (89%)	30 (86%)	32 (91%)
Qwen3-8B	C keeps	18	18 (100%)	17 (94%)	17 (94%)
	C switches	17	17 (100%)	16 (94%)	17 (100%)
	none	35	34 (97%)	31 (89%)	35 (100%)
Llama-3.2-3B	C keeps	18	18 (100%)	12 (67%)	13 (72%)
	C switches	17	16 (94%)	10 (59%)	11 (65%)
	none	35	32 (91%)	18 (51%)	25 (71%)

models (Table 6.4). The gap lies in the endpoint's specification, not in the models. It is reported as pre-registered and the endpoint is not changed. A post-hoc sensitivity check, not pre-registered, that allows the trailing symbol restores agreement with strict on all 36 Qwen3-8B cases; the only other Qwen3-8B disagreement is PB065-E.

6.5 Pilot-B: resources

Table 6.5 gives the stored median token counts and latencies (pre-registered endpoint 6). R prompts are longer than E for every model (123 vs 109 input tokens for Qwen, 138 vs 124 for Llama). The input count covers the whole prompt, including the chat template and the fixed instruction. Llama's R outputs are also longer. These are correlates, not causes. Section 6.10 examines question-level token counts in a post-hoc exploratory analysis.

6.6 Pilot-B: output audit of the primary model

Method. Two AI models, Claude and GPT, pre-tagged all 210 Qwen3-4B Pilot-B outputs under written audit instructions, each blind to the other's tags. Each output received an error category, the first applicable in a pre-registered order, and an English-compliance tag. The taggers agreed on the category for 206/210 outputs (Cohen's $\kappa = 0.94$) and on

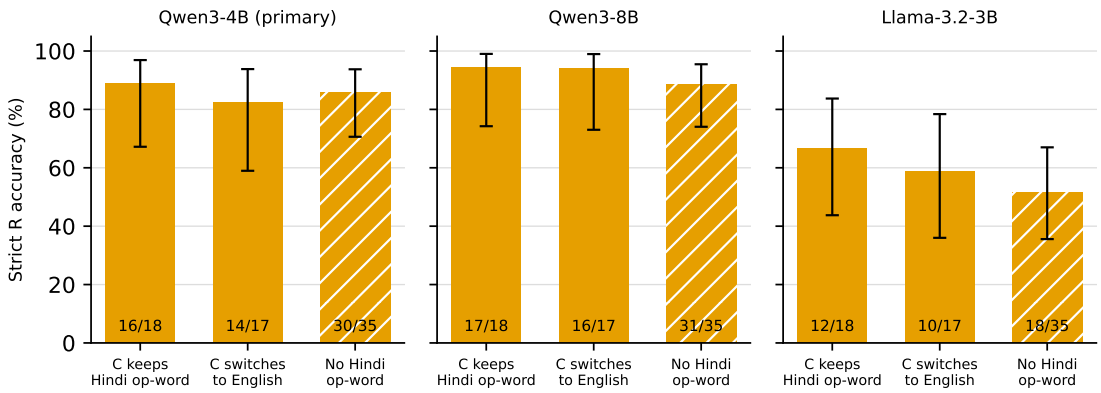


Figure 6.2: Pilot-B strict R accuracy by operation-word stratum with 95% Wilson intervals; bar labels give correct/families. Regenerated from `results/pilot_b/*/per_rendering.csv` with the stratum rule of `results/pilot_b/strata.py`.

Table 6.4: Pilot-B secondary endpoint `numeric_literal_v1` beside strict accuracy (%). The Qwen shortfall is the trailing-₹ specification gap.

	Strict			Numeric		
Backbone	E	R	C	E	R	C
Qwen3-4B	94.3	85.7	94.3	72.9	82.9	82.9
Qwen3-8B	98.6	91.4	98.6	84.3	74.3	80.0
Llama-3.2-3B	94.3	57.1	70.0	94.3	57.1	70.0

English compliance for 208/210 ($\kappa = 0.96$). A human then checked the 4 disagreement rows and 20 seeded random agreed rows (seed 101), with the taggers anonymized as A and B. The human label agreed with tagger A on 22/24 and with tagger B on 21/24. The final label is the human label where checked (24 outputs) and the agreed AI label otherwise (186 outputs). The audit is therefore an *AI pre-tag, human-checked on disagreements and a sample*. One human label, "ambiguous" (PB033-R, flagged as an ambiguous question), is mapped to *unresolved*. This is an audit of model outputs, not of the dataset.

Findings. Of Qwen3-4B's 18 strict failures, 6 are format-only. The substantive errors concentrate in R: 8, against 1 in E and 3 in C. English explanations fell from 70/70 on E to 53/70 on R and 59/70 on C (Table 6.6). The *language_drift* rows are strict-correct outputs (with *no_error* they sum to the strict-correct counts 66/60/66) whose explanation drifted away from English, so instruction following erodes on R and C even when the answer is right.

Exploratory (Llama; not pre-registered). A heuristic count found that Llama explained 60/70 R and 27/70 C outputs in Romanized-Hindi prose (E: 0/70; no Devanagari). On C, Hindi-prose explanations were correct 13/27 (48%) against 36/43 (84%) for English explanations; on R, 35/60 against 5/10. This association is observational and confounded

Table 6.5: Pilot-B median input tokens, output tokens and generation latency (s) by form (70 renderings each; latency excludes model load and tokenization).

	Input tokens			Output tokens			Latency (s)		
Backbone	E	R	C	E	R	C	E	R	C
Qwen3-4B	109	123	119	104	98.5	93.5	9.59	9.26	8.65
Qwen3-8B	109	123	119	72.5	73.5	72.5	7.33	7.32	7.40
Llama-3.2-3B	124	138	134.5	99	119.5	96.5	5.95	7.14	5.75

Table 6.6: Primary audit category per Qwen3-4B Pilot-B output (70 per form) and English-compliant explanations.

Category	E	R	C
no_error	66	47	56
format_only	3	2	1
language_drift	0	13	10
wrong_relation_operation	1	4	2
misread_quantity	0	2	0
arithmetic_slip	0	1	1
unresolved	0	1	0
English-compliant explanation	70	53	59

with item difficulty, since harder items may both trigger Hindi explanations and cause errors.

6.7 Post-hoc audit extension: Qwen3-8B and Llama-3.2-3B

Evidence label: post hoc, not pre-registered (28 October–3 November 2026). AI pre-tags by three taggers (Claude Opus, GPT, Claude Sonnet), human-checked on every disagreement and a seeded random sample (returned 10 November 2026).

The pre-registered audit covered only the primary model. The same written audit instructions were then applied to all 210 Pilot-B outputs of each secondary backbone by three taggers, each blind to the others: A = Claude Opus, B = GPT and C = Claude Sonnet. C was added while GPT was unavailable; B was added once it was available again. The cross-family pair A–B is the headline, as in the primary audit. Table 6.7 gives the agreement. The two Claude taggers agree with each other more than either agrees with GPT, which is the pattern expected if same-family taggers share blind spots, and is why the cross-family comparison is preferred.

A human-check sheet was built with every row on which any two taggers differ (13 for Qwen3-8B, 24 for Llama) plus 30 seeded random rows per backbone on which all three

Table 6.7: Pairwise tagger agreement in the post-hoc audit extension (210 Pilot-B outputs per backbone; agreements out of 210, Cohen's κ in parentheses; A = Claude Opus, B = GPT, C = Claude Sonnet).

Backbone	Label	A–B	A–C	B–C
Qwen3-8B	Error category	200 (0.871)	207 (0.959)	197 (0.830)
	English compliance	209 (0.985)	209 (0.985)	208 (0.971)
Llama-3.2-3B	Error category	187 (0.842)	205 (0.965)	190 (0.862)
	English compliance	208 (0.981)	210 (1.000)	208 (0.981)

Table 6.8: Final audit labels for the secondary backbones on Pilot-B (70 outputs per form; human label on 97 checked rows, otherwise the unanimous AI label). Descriptive and post hoc; strict scores are unchanged.

	Qwen3-8B			Llama-3.2-3B		
Category	E	R	C	E	R	C
no_error	69	37	61	66	0	33
format_only	1	0	0	1	0	2
language_drift	0	27	8	0	40	16
wrong_relation_operation	0	2	1	2	12	9
misread_quantity	0	4	0	0	10	7
arithmetic_slip	0	0	0	0	3	1
unit_error	0	0	0	1	0	0
truncated	0	0	0	0	5	2
English-compliant explanation	70	37	62	70	0	36

agree, 97 rows in all. It was completed by a bilingual human checker [*checker's role: team member or external volunteer; credit preference*]. All 97 rows were labelled with a note. The checker agreed with all 60 unanimous sample rows. On the 37 disagreement rows the checker's category matched tagger A (Claude Opus) on 32, tagger B (GPT) on 5 and tagger C (Claude Sonnet) on 30. Over all checked rows, category agreement with the human was A 92/97, B 65/97 and C 90/97; English-status agreement was 94, 95 and 95 of 97. Final labels are the human label where checked and the unanimous AI label otherwise (`results/pilot_b/audit/final_audit_<model>.csv`). Table 6.8 gives them.

Two patterns stand out. First, for both backbones the strict failures on R are substantive, not format-only: there are no format-only R failures, while Qwen3-8B has 6 substantive R errors and Llama 25 substantive R errors plus 5 truncations. Second, language drift is large: Llama wrote no R explanation in English (0/70), and Qwen3-8B only 37/70. Language drift is assigned only when no earlier category applies, so it often co-occurs with correct answers. The heuristic Llama language counts in the previous section are separate from this extension.

Table 6.9: Original cohort strict accuracy (24 families per form) with 95% Wilson intervals. No strict-invalid outputs.

Backbone	E		R		C	
Qwen3-4B (primary)	24/24	100.0 [86.2, 100.0]	22/24	91.7 [74.2, 97.7]	24/24	100.0 [86.2, 100.0]
Qwen3-8B	24/24	100.0 [86.2, 100.0]	23/24	95.8 [79.8, 99.3]	24/24	100.0 [86.2, 100.0]
Llama-3.2-3B	21/24	87.5 [69.0, 95.7]	15/24	62.5 [42.7, 78.8]	22/24	91.7 [74.2, 97.7]

Table 6.10: Original cohort paired differences (pp) with 95% family-bootstrap intervals (10,000 resamples, seed 101) and discordant counts. Zero-width intervals at ceiling are not evidence of equivalence.

Backbone	E−R	E−C	R−C
Qwen3-4B	+8.3 [0.0, +20.8] (2/0)	0.0 [0.0, 0.0] (0/0)	−8.3 [−20.8, 0.0] (0/2)
Qwen3-8B	+4.2 [0.0, +12.5] (1/0)	0.0 [0.0, 0.0] (0/0)	−4.2 [−12.5, 0.0] (0/1)
Llama-3.2-3B	+25.0 [+4.2, +45.8] (7/1)	−4.2 [−16.7, +8.3] (1/2)	−29.2 [−45.8, −12.5] (0/7)

6.8 Original cohort results

Tables 6.9 and 6.10 and Figure 6.3 report the 24-family original cohort. All runs completed (72/72) with zero strict-invalid outputs, and the numeric endpoint agrees with strict scoring on all 72 renderings for each model.

Individual errors. Qwen3-4B made two R errors. In PIL001-R it answered 41 against gold 31, treating *Riya se 5 kam* ("5 fewer than Riya") as addition; this is an isolated operation-word (*kam*) misreading. In PIL028-R it answered 50/9 cm, a wrong value. Qwen3-8B's only error is PIL025-R (answered 3, a wrong value). Llama made 14 errors, 9 of them in R; PIL007 is wrong in all three forms (answered 12), so it is not a language effect.

6.9 Descriptive cross-cohort comparison

Table 6.11 is the one pre-specified side-by-side comparison. It reports each cohort's own within-cohort estimates; there is no pooling, no cross-cohort test, no combined interval and no cross-cohort pairing. The cohorts differ in families, sample size, topic and operation-word composition, review process and C definition, so no difference between them can be attributed to any single factor.

The direction replicates for every model: R is the lowest form in both cohorts. Qwen is near ceiling in the original cohort (E = C = 100%). Llama's R deficit appears in both cohorts (E−R +37.1 and +25.0 points). C's position for Llama differs: well below E in Pilot-B, but at E's level in the original cohort. C's position relative to E is therefore model- and cohort-dependent.

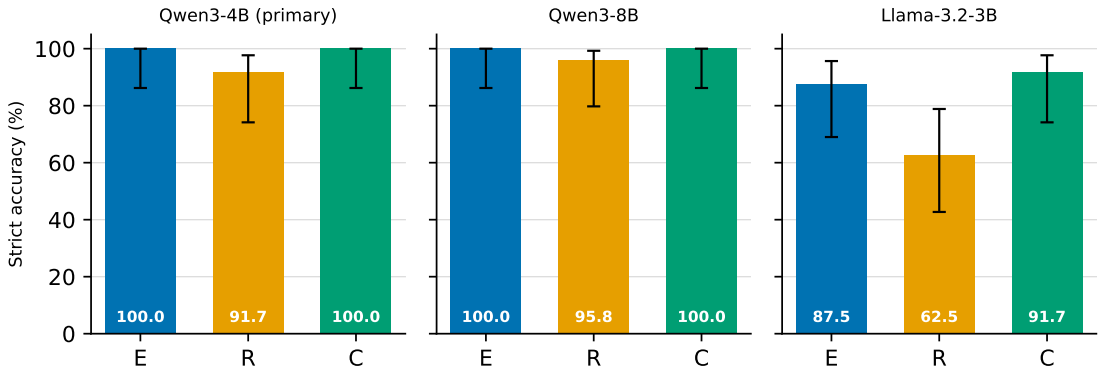


Figure 6.3: Original cohort strict accuracy by form with 95% Wilson intervals (24 families per bar). Regenerated from `results/pilot_original/*/per_rendering.csv`.

Table 6.11: Pre-specified descriptive comparison of the two cohorts (never pooled). Strict accuracy (%) and paired differences (pp), each computed within its own cohort.

Backbone	Cohort	N	E	R	C	E−R	E−C	R−C
Qwen3-4B	Pilot-B	70	94.3	85.7	94.3	+8.6	0.0	−8.6
	Original	24	100.0	91.7	100.0	+8.3	0.0	−8.3
Qwen3-8B	Pilot-B	70	98.6	91.4	98.6	+7.1	0.0	−7.1
	Original	24	100.0	95.8	100.0	+4.2	0.0	−4.2
Llama-3.2-3B	Pilot-B	70	94.3	57.1	70.0	+37.1	+24.3	−12.9
	Original	24	87.5	62.5	91.7	+25.0	−4.2	−29.2

6.10 Post-hoc exploratory analysis: tokenization and question length

Evidence label: POST-HOC EXPLORATORY. Written on 27 October 2026, after every Pilot-B and original-cohort output had been seen, and not frozen before the outputs were seen. Descriptive only: the cohorts are small (70 and 24 families), there is no multiplicity correction and no causal claim is made. The two cohorts are reported separately and never pooled.

Method. Question texts (the frozen renderings of every accepted family, without the chat template or instruction) were tokenized with the pinned Qwen3-4B tokenizer (`tokenizers` 0.23.3, no special tokens); Qwen3-8B uses the same tokenizer. As a check, for both Qwen models the stored whole-prompt R−E and C−E input-token differences equal the question-only differences in 70/70 and 24/24 families. The Llama tokenizer is gated and was not downloaded, so for Llama the Qwen3 counts serve as a model-independent length proxy, alongside the R−E differences of Llama's own stored whole-prompt counts. Operation words were matched whole-word against the lexicon of Section 3.6, and "pieces" counts the tokens overlapping each word. Correctness is the strict endpoint. Among fami-

Table 6.12: Question-only tokenization per form with the Qwen3 tokenizer (post-hoc exploratory). Medians per family; the last column is the share of alphabetic words split into three or more pieces.

Cohort	Form	Tokens	Words	Tokens/word	Words ≥3 pieces
Pilot-B	E	35	24	1.45	1.4%
	R	49	24	2.00	16.5%
	C	45	24.5	1.78	9.4%
Original	E	37.5	28.5	1.33	0.2%
	R	59.5	30	1.96	15.0%
	C	44	29.5	1.54	3.4%

Table 6.13: Families with E correct: R−E question tokens (Qwen3 counts; median [IQR]) for R-wrong vs R-correct families, exact two-sided permutation p, and logistic odds ratio for R correct per +10 tokens [Wald 95%]. Post-hoc exploratory; no multiplicity correction.

Cohort	Backbone	R wrong: n, R−E	R correct: n, R−E	p	OR [95%]
Pilot-B	Qwen3-4B	9, 15 [9, 16]	57, 14 [10, 19]	0.17	2.49 [0.67, 9.22]
	Qwen3-8B	6, 13 [9.25, 19.75]	63, 15 [11, 18.5]	0.93	1.09 [0.31, 3.83]
	Llama-3.2-3B	28, 16.5 [13.25, 21]	38, 13.5 [10, 16.75]	0.073	0.48 [0.21, 1.09]
Original	Qwen3-4B	2, 10.5	22, 25.5	0.14	not fitted
	Qwen3-8B	1, 30	23, 21	0.83	not fitted
	Llama-3.2-3B	7, 24 [12, 31]	14, 19 [12.25, 27.75]	0.85	0.93 [0.46, 1.85]

lies with E correct, R-wrong and R-correct families were compared with an exact two-sided permutation test of the mean R−E token difference, and a one-predictor logistic regression was fitted only when both classes had at least 5 families and there was no separation. The analysis script is deterministic (`analysis/tokenization/`).

Tokenization per form. Table 6.12 shows that R questions are longer than E at about the same word count: the extra cost is in segmentation, not in more words. Per family, R−E is +15 tokens (median; IQR 10.25–18.75; ratio 1.42; R longer in 70/70) in Pilot-B and +22.5 (IQR 12.75–31.75; ratio 1.61; 24/24) in the original cohort, with R−E tokens per word +0.60 in both. C−E is +8 tokens in both cohorts (ratio 1.24 and 1.21; C longer in 67/70 and 20/24). Hindi operation words are only slightly more fragmented than other R words: 54 occurrences in 37 Pilot-B R renderings average 2.07 pieces (27.8% with three or more) and 63 occurrences in 21 original R renderings average 2.21 (39.7%), against 1.76 and 1.78 pieces for the average alphabetic R word.

Length and correctness. For the Qwen models, R-wrong families were not longer (Table 6.13); an odds ratio above 1 means that a longer R−E difference went with R being correct more often, not less. For Llama in Pilot-B the R-wrong families had larger R−E differences (16.5 vs 13.5 tokens, $p = 0.073$), and the longest tercile of R questions (54–87 tokens) had the lowest R accuracy, 43.5% against 60.9% and 66.7%, although Llama's

E accuracy in that tercile was 95.7%. Llama's own stored whole-prompt counts give the same picture (Pilot-B medians 16 vs 13, $p = 0.074$; original 24 vs 18.5, $p = 0.83$). Raw R length for Pilot-B Llama gives $p = 0.037$ (median 53 vs 47), the only nominal p below 0.05 among the 14 post-hoc permutation tests (plus 6 logistic fits). Only Llama writes noticeably longer outputs on R (Table 6.5).

Limits. The Qwen tests rest on 1–9 R-wrong families per cohort and have almost no power, so a null here is not evidence that length has no effect. Length is confounded with item content, the number of quantities and the number of Hindi words, and fragmentation with how rare the Romanized register is in pretraining. Llama piece counts are unavailable, the lexicon is an unconfirmed proposal matched by surface form only, and only the strict endpoint was used. The interpretation is given in Section 7.6.

Chapter 7

Discussion

7.1 Romanized Hindi, not code-mixing, is the hard case here

In both cohorts and for all three backbones, the Hinglish rendering was never the lowest-scoring form; Romanized Hindi was. For both Qwen models, Hinglish matched English in observed aggregate accuracy in both cohorts. Popular intuition often treats mixing two languages in one sentence as the difficult part of Hinglish. These results suggest that, for these models and items, the difficulty sits more with the Hindi-dominant Roman-script form. Insertional Hinglish keeps many English content words (*ratio*, *add*, *total*), and these may anchor the model's understanding.

Some care is needed in stating this. "E = C" here is equality of two observed proportions on 70 or 24 families. It is not itemwise identity (there are 3/3 discordant families for Qwen3-4B in Pilot-B) and not an equivalence test with a margin. For Llama in Pilot-B, C is 24.3 points below E, so "code-mixing is harmless" is not a supported general claim. Because all three forms are in Latin script, the design cannot separate a romanization penalty from a Hindi-vocabulary or register effect (Chapter 10).

7.2 Model dependence

The size of the R deficit varies roughly four- to fivefold across backbones (7–9 points for Qwen, 37 for Llama in Pilot-B). Llama is as good as Qwen3-4B on English (94.3% each) but far worse on R. A benchmark result on one model family is therefore a weak guide to another, and the development screen showed that even *whether* a model can be scored under a given format depends on the model.

7.3 Scale

Within Qwen3, the 8B model improves every form, but the R deficit remains (8.6 $\rightarrow$ 7.1 points in Pilot-B). Scale helps overall accuracy more than it closes the language-form gap. In the development screen the most dramatic scale effect was on instruction following, not mathematics: 1.7B could not follow the answer format at all, while 4B and 8B followed it perfectly. With one model family per size, this is an observation, not a scaling law.

7.4 The operation-word null result

The development study produced a vivid hypothesis: R errors come from ambiguous Hindi operation words such as *bhinn* (fraction/different). The keep/switch design of Pilot-B was built to test it, and the strata show no concentration of R errors in families containing such words. R accuracy is similar whether C keeps the word, C switches it, or there is no such word at all. The audit points the same way: Qwen3-4B's eight substantive R errors are spread across wrong relations, misread quantities, an arithmetic slip and one unresolved case, and Llama's R errors include plain arithmetic slips. One clear operation-word misreading did occur in the original cohort (*Riya se 5 kam* read as addition), so such misreadings happen. They do not, however, explain most of the gap. With strata of 17–35 families this is a descriptive null, not proof that vocabulary plays no role.

This null result illustrates why the protocol separated development evidence from cohort evidence. Two striking fixture errors suggested a mechanism, and a balanced, pre-registered design did not support it.

7.5 Instruction-following erosion

A second, quieter effect runs through the data: models increasingly ignore the instruction to explain in English as the input moves away from English. Under prompt v1, 8 of 12 R fixture explanations were in Devanagari. Under v2, in Pilot-B, Qwen3-4B's English compliance fell from 70/70 (E) to 53/70 (R) and 59/70 (C). Llama explained 60 of 70 R outputs in Romanized-Hindi prose. The model often reasons in the language it was addressed in, even when told otherwise. This matters for applications that need consistent output language, and it matters for evaluation, because a scorer or extractor tuned to English text can silently penalize such outputs.

7.6 Tokenization length: at most a minor correlate

Evidence label: this section interprets the POST-HOC EXPLORATORY analysis of Section 6.10; it is not a pre-registered result.

In a post-hoc exploratory analysis with the pinned Qwen3 tokenizer, Romanized-Hindi (R) questions were consistently longer than their English (E) counterparts: a median of $+15$ tokens (ratio 1.42) in Pilot-B and $+22.5$ (ratio 1.61) in the original cohort, at nearly the same word count, because R words split into more pieces (2.0 vs 1.45 tokens per word). Hindi operation words were only slightly more fragmented than other R words. Length does not account for the R deficit in the Qwen models. Their R errors did not occur in longer or more fragmented items: the median R$-$E difference was 15 vs 14 tokens for

Qwen3-4B and 13 vs 15 for Qwen3-8B. In addition, code-mixed (C) questions were also longer than E (median +8 tokens), yet Qwen's C accuracy equalled its E accuracy in both cohorts. For Llama-3.2-3B there was a weak association in Pilot-B: R-wrong families had larger R−E differences (16.5 vs 13.5 tokens; exact $p = 0.07$), and the longest tercile had the lowest R accuracy. This association did not appear in the original cohort and would not survive any correction for multiple testing. Given the small samples, the many post-hoc comparisons and the confounding of length with item content and register rarity, this report treats tokenization length as at most a minor correlate. The R gap is interpreted as a language-form effect, not a demonstrated tokenization effect. Interventions that hold the content fixed while changing the wording (lexical swaps, length padding) would be needed to test the mechanism.

7.7 Comparison with prior work

The direction of the main result matches CodeMixTax [1], which reports Romanized Hindi below Romanized Hinglish across models on translated GSM8K. The gaps there are larger (for example 90.26/66.18/40.63 for Gemma-3-4B on E/Romanized-Hinglish/Romanized-Hindi). That study uses scheme-style romanization of machine-translated items, while this project uses newly drafted, human-reviewed renderings in natural spelling, with a strict scorer checked on excluded data. A replication of the direction under these different conditions makes the ordering more credible. Differences in magnitude cannot be attributed to any one factor, because the models, items and protocols all differ.

The scoring findings echo Peter et al. [2]: evaluation details can create apparent language effects. Here, a single phrase in the answer-format instruction produced form-dependent scoring failures on development data. The pre-registered secondary endpoint's trailing-₹ gap shows that even a carefully specified extractor can have blind spots, and the unlisted English unit phrases (`11 years old`) show that strict scoring has a cost too. Reporting both endpoints side by side, unchanged, lets readers see these effects instead of having them silently absorbed into accuracy.

Sarvam-M [13] and the health-triage study of Khullar et al. [14] report degradation on Romanized Indic input relative to native script; this project cannot test that contrast because it has no Devanagari arm. CodeMixBench [11] reports lower accuracy on mixed-script Hindi–English GSM8K than on English, which is consistent with Llama's Pilot-B E−C gap but not with Qwen's E = C. This is a reminder that "code-mixed" covers several scripts and registers.

Chapter 8

Phase 2: Design and Feasibility

This chapter records the design, feasibility and planning work completed before any Phase-2 training output. It is kept as written at the time, so its numbers are planning evidence; the executed study and its results are in Chapter 9.

8.1 The training study

Design. The essential comparison (D001) is three fine-tuning arms on matched mathematical content with identical English solution targets and matched exposure:

- **E+E**: each TRAIN family supervises the English solution twice, from the E question;
- **E+R**: once from the E question and once from the R question;
- **E+C**: once from the E question and once from the C question.

Training uses the frozen TRAIN split (160 families), with DEV (40 families) for sanity checks only. A single confirmatory evaluation then runs on the untouched FINAL_TEST (100 families) to answer RQ2 and RQ3. The primary estimand (D040) is Δ_C: the E+C minus E+R difference in strict accuracy on C questions, averaged over the 100 FINAL_TEST families and three paired seeds. English retention (E+C vs E+E on E), R transfer (E+R vs E+E on R) and specificity ($\Delta_C - \Delta_R$) are descriptive secondaries, and the unchanged Qwen3-4B on FINAL_TEST is the baseline. Matching exposure does not match input tokens or runtime, so both were measured per arm.

Backbone. Qwen3-4B stays the backbone: it follows prompt v2 cleanly, shows the R deficit the study targets, its stack is verified on the T4, and it is the backbone of the submitted paper. Llama-3.2-3B, which has more C headroom, is kept only as an optional second backbone if the ceiling flag below fired (D042); it did not fire (D049).

8.1.1 Training smoke test (D021 gate): GO

Evidence label: engineering test only (28 October 2026). 16 TRAIN families (D041); the adapters were discarded and never evaluated, and DEV and FINAL_TEST were not accessed.

Decision D021 fixed the go/no-go thresholds before any measurement: median $\leq$56 s per update stays on Colab; 56–150 s moves to Kaggle 2$\times$T4; $>$150 s falls back to a smaller model. The smoke test ran all three arms of Qwen3-4B QLoRA on a Colab T4 with a project-owned training loop and self-built completion-only labels, on exactly the D021

Table 8.1: D021 training smoke test on a Colab T4 (Qwen3-4B QLoRA; median over updates 3–6). Engineering evidence only.

Arm	Median s/update	Peak allocated	Min free	Scaler skips	Checkpoint reload
E+E	19.8	5.02 GB	10.1 GB	0	exact (probe diff 0.0)
E+R	20.4	5.03 GB	10.1 GB	0	exact (probe diff 0.0)
E+C	20.6	5.03 GB	10.1 GB	0	exact (probe diff 0.0)

stack (torch 2.11.0+cu130, transformers 4.51.3, bitsandbytes 0.50.2, PEFT 0.15.2). The verdict was **GO: stay on the Colab T4**, with no failed checks (Table 8.1). All 20 included files of the results ZIP match its checksum list.

A longer 32-update E+C probe ran at a median 19.8 s per update with no out-of-memory error, finite losses and no scaler skips. Every update supervised the same 732–814 English target tokens in all three arms, from inputs of 2.6–2.8k tokens per 16-example update. The timing is slightly pessimistic, because updates 5–6 run in a fresh process and include start-up cost, and it holds for examples of at most about 207 tokens.

Memorisation finding. Training loss fell from about 2.55 to 0.32–0.41 within six updates, and in the 32-update probe it reached about 0.00 by update 24. The 16 smoke families (32 examples, two updates per pass) were memorised within about 12 passes. The provisional 256 updates over 160 families (D015) would be about 13 passes, roughly 1.5 h per arm and seed at 20.6 s per update, and so would sit in the same memorisation regime. Decision D042 therefore kept training on the free Colab T4 with Qwen3-4B (so the $>$150 s fallback no longer applies), handled the C ceiling through the DEV ceiling flag rather than a backbone switch, and required the update count to be fixed before any training.

8.1.2 Training schedule (D043)

Decision D043 (3 November 2026) fixed the schedule before any Phase-2 training or FINAL_TEST output:

- all 160 TRAIN families (`train_A.v2` + `train_B.v2`); 320 examples per arm; 16 examples per update, so 20 updates per pass;
- **3 passes = 60 updates** per run, replacing the provisional 256;
- AdamW, learning rate 10^{-4}, 6 linear warm-up updates, then cosine decay to 10%; otherwise the D021 recipe is unchanged: LoRA rank 16 and alpha 32 on all seven linear projections (33,030,144 trainable parameters), microbatch 1 with 16-step accumulation, a 1,024-token sequence cap, completion-only loss, NF4 with double quantization, FP16 compute and gradient checkpointing;

Table 8.2: Simulated probability that the 95% lower bound of Δ_C exceeds 0 ($N = 100$ FINAL_TEST families, 3 seeds). Brackets: median 95% half-width (pp) under no effect. Planning evidence from Pilot-B outcomes of unchanged models; gain-only effects, so optimistic.

p_0 (E+R on C)	q_0	$\delta = 0$ (false support)		$+3$ pp	$+5$ pp	$+10$ pp
0.943	0.035	0.016	[2.1]	0.56	0.89	impossible
0.943	0.071	0.023	[2.9]	0.51	0.96	impossible
0.857	0.071	0.025	[2.9]	0.42	0.78	1.00
0.857	0.114	0.027	[3.8]	0.31	0.67	1.00
0.857	0.200	0.023	[5.0]	0.22	0.56	1.00
0.70	0.071	0.020	[3.0]	0.41	0.74	0.995
0.70	0.200	0.025	[5.0]	0.21	0.45	0.95

- seeds 11, 22 and 33; within a seed, family order, example order and LoRA initialization are identical across the three arms (paired);
- the checkpoint after update 60 is evaluated; DEV is used only for an unchanged-model ceiling check and a post-training sanity report, never for selection.

The expected cost is about 21 minutes per run for nine runs (three arms $\times$ three seeds). The confirmatory evaluation covers ten checkpoints (nine trained plus the unchanged model) $\times$ three forms $\times$ 100 families = 3,000 generations, roughly 6–7 T4-hours at Pilot-B latency. FINAL_TEST is evaluated once, only after all nine checkpoints are frozen and hashed.

8.1.3 Power, precision and decision rules (D040)

Evidence label: planning evidence. A Monte Carlo simulation that reads only the stored Pilot-B outcomes of the unchanged models; no training, new inference or FINAL_TEST inspection. Its effect model allows gains only, so it is optimistic. The direction was adopted (D040) and the full protocol was frozen as `phase2/PREREGISTRATION.md` (D044, 3 November 2026), committed before any Phase-2 training, DEV or FINAL_TEST output.

The simulation models each family as solid, fragile or hard, calibrated so that the E+R arm's C accuracy is p_0 and the discordance between two seeds of one arm is q_0, and lets E+C raise accuracy by a mean of δ. Each cell uses 4,000 simulated datasets, and the Wald decision interval used for speed agreed with a percentile family bootstrap (power 0.717 vs 0.710, decision agreement 98.7%; and 0.967 vs 0.967, 100%). Table 8.2 gives selected cells for $N = 100$ families and three seeds.

What 100 families can support. Δ_C can be estimated to about $\pm$2–5 points (95% half-width; $\pm$3–4 in the central cases). A $+3$-point gain is detected with only about 21–58% probability, and the approximate 80%-power minimum detectable effect is 3–7 points. A "useful gain" claim (lower bound above $+5$ points) is essentially unreachable: its probability is at most 0.025 at a true $+5$ and 0.41–0.60 at a true $+10$. The ceiling is the main risk: unchanged Qwen3-4B already scores 94.3% on Pilot-B C, so if the E+R arm stays there the

largest possible gain is 5.7 points, and $+10$ is impossible. English non-inferiority within 2 points is shown with probability only 0.52 or 0.29 even when nothing is lost. More families would help (central case, $+3$ points: 0.31/0.55/0.73/0.96 at $N = 100/200/300/600$), but only 100 FINAL_TEST families exist, and adding DEV would break the DEV firewall.

Decision labels (primary Δ_C only). *Support*: 95% lower bound > 0. *Useful support*: lower bound $> +5$ points. *Informative null*: the interval contains 0 and its upper bound is $< +5$ points. *Harm*: upper bound < 0. Otherwise *inconclusive*. The primary result is therefore framed as a pre-registered estimate with a precision target, not a significance hunt. English retention uses a -2-point margin and is reported as "not established" unless the lower bound exceeds -2 points.

Other adopted elements (D040). Strict scorer v1.0.0 stays primary. `numeric_literal_v2`, which also accepts ₹, Rs, INR or "rupees" when the gold unit is rupees, is pre-registered as a new secondary only; it closes the trailing-₹ gap of Table 6.4 without changing the cohort results. The unchanged model is run on DEV before training, with a ceiling flag if its C accuracy is at least 95%. A self-translate baseline is the optional control, and no Devanagari arm is planned (cost, new reviewed renderings and a new FINAL_TEST version).

Chapter 9

Phase 2: Training Methods and Confirmatory Results

Evidence label: completed pre-registered training study. Strict-accuracy Δ_C is the sole confirmatory estimand. The registered decision is *Harm*: -4.33 percentage points (pp), with a 95% paired family-bootstrap interval of $[-8.00, -0.67]$. All secondary endpoints, retention flags, per-seed comparisons and strata are descriptive. The protocol includes Amendments 1–3; the access deviation and execution incidents are disclosed below.

9.1 Methods

Question, data and arms. The study asks whether training on Hinglish rather than Romanized Hindi improves accuracy on Hinglish mathematical word problems. **E**, **R** and **C** denote English, Romanized Hindi with frozen borrowings, and insertional Roman-script Hinglish. TRAIN contains 160 Claude-drafted families; DEV contains 40 GPT-drafted families used only for sanity checks; FINAL_TEST contains 100 GPT-drafted families. Each family has one frozen rendering per form. Pilot, Pilot-B and FMT2 families are excluded. The earlier smoke-test adapters were discarded.

Each TRAIN family supplies two examples, giving 320 examples per arm: E+E repeats E; E+R pairs E with R; E+C pairs E with C. Every form supervises the identical English `reference_solution_en` target plus `<|im_end|>`. Thus the arms match mathematical content, example exposure and targets. The labels mask the prompt and empty think block; target-token equality across forms is asserted before training. No packing or truncation is allowed, and sequences must fit 1,024 tokens. Data hashes are checked before use; a mismatch halts execution. The evaluated test file is `phase2/final_test.v3.json`, SHA-256 `da89dd9a626e438dffca39e3a74c4c8abac6bb57501495723b87e6ee143f3d46`.

Backbone and training recipe. Qwen3-4B is the sole registered backbone, with model and tokenizer revision `1cfa9a7208912126459214e8b04321603b3df60c`. Training uses a Colab T4 and the pinned stack: torch 2.11.0+cu130, transformers 4.51.3, bitsandbytes 0.50.2, PEFT 0.15.2 and accelerate 1.6.0. QLoRA uses double-quantized NF4 base weights, FP16 compute, FP32 adapters, rank 16, alpha 32 and dropout 0.05 on q/k/v/o/gate/up/down projections. Attention uses SDPA math; gradient checkpointing is non-reentrant and caching is disabled.

Microbatch size is 1 with accumulation over 16 examples, or eight families per update. Loss averages supervised tokens within equally weighted examples. AdamW uses zero weight decay and gradient clipping at 1.0. Three passes produce exactly 60 successful updates. The learning rate rises linearly to 10^{-4} over six updates, then decays by a cosine schedule to 10^{-5} at update 60. Only that final checkpoint is evaluated. This schedule was a resource-feasible choice made before Phase-2 outputs, not a validated optimum.

Seeds 11, 22 and 33 pair initial adapter tensors and family exposure order across arms, but do not pair dropout draws. Within each seed and pass, families are sorted by the SHA-256 of UTF-8 `phase2-D043:{s}:{p}:{family_id}`, using decimal seed/pass and family-ID tie-breaking; E precedes the arm-specific second example. Exposure lists were frozen and hashed. Training follows seed-slot order, with EC, ER and EE within each slot.

Evaluation and DEV firewall. The unchanged model and each adapter use the frozen `test1-english-final-v2` prompt, strict scorer v1.0.0, non-thinking greedy decoding, fresh contexts, batch size 1 and evaluation-engine seed 11. Context and output caps are 2,048 and 1,024 tokens. Adapters are evaluated unmerged on the same NF4 base with FP16 compute. Invalid, missing-answer, refused, ambiguous and truncated responses score zero; an absent generation remains unresolved. Numeric-literal v1 and v2 are descriptive endpoints. Under Amendment 1, v2 keeps v1 unless v1 returns `unsupported_answer`; only then does it try the registered rupee-specific extensions. Non-rupee gold retains v1.

A validated unchanged-model DEV archive was required before training. Neither adapter DEV accuracy nor loss curves could select a checkpoint, alter the recipe or trigger a retry. The high unchanged-DEV accuracy flag was a ceiling-risk proxy, with threshold C $\geq 38/40$; it did not change the primary estimand or turn S2 into a confirmatory endpoint. No self-translation control or second backbone belongs to this registration.

Estimand and inference. Let $y(i, s, a, l)$ be binary strict correctness for family i, seed s, training arm a and evaluation form l. The sole confirmatory contrast is

$$d_i = \frac{1}{3} \sum_{s \in \{11,22,33\}} [y(i, s, EC, C) - y(i, s, ER, C)], \qquad \Delta_C = \frac{1}{100} \sum_{i=1}^{100} d_i.$$

Families and seeds receive equal weight. This is the effect of the complete training-rendering recipe, not an isolated effect of code-switching. The observed mean describes the finite test set; uncertainty intervals approximate sampling of independent families from the fixed authoring and review process, conditional on the trained checkpoints. They do not cover training-seed variance.

The paired family bootstrap draws 10,000 resamples of 100 families, retaining each family's full arm/form/seed record. Sorted family IDs and a fresh Python

`random.Random(101)` determine a shared index matrix, reused across endpoints, contrasts and per-seed intervals. Percentile bounds use linear quantiles. Companion Wald and bounded Hoeffding intervals are reported for the primary contrast. Checkpoint accuracies receive Wilson 95% intervals over families, never over pooled seed rows. Crossed family–seed resampling is excluded by Amendment 1.

For unrounded strict-accuracy bounds $[L, U]$ in pp, *Useful support* requires $L > 5$; *Support*, $0 < L \leq 5$; *Informative null*, $L \leq 0 \leq U < 5$; *Harm*, $U < 0$; and *Inconclusive*, $L \leq 0$ and $U \geq 5$. Identical family contrasts or a zero-width interval override these with *inconclusive—degenerate empirical interval*. All secondaries and per-seed exact McNemar tests are descriptive, without Holm testing.

9.2 Execution and integrity

9.2.1 Completed training and adapter freeze

All nine training runs were valid on attempt 1: each completed 60/60 successful updates, had zero scaler skips, and was not resumed. There were no training infrastructure retries and no reserve seeds. Initial adapter and data-order pairing checks passed for each seed. Table 9.1 reproduces the training log; its DEV scores were sanity checks only.

Table 9.1: Retained training runs, in execution order. DEV counts are out of 40 per form; invalid counts cover all DEV forms. Tokens are mean output lengths. Every row is valid for the gate.

Run	Attempt	Steps; skips	s/update	DEV E/R/C	Invalid	Tokens
EC-s11	1	60/60; 0	20.66	24 / 13 / 19	23/120	43
ER-s11	1	60/60; 0	19.18	23 / 12 / 17	25/120	44
EE-s11	1	60/60; 0	20.59	25 / 15 / 17	23/120	46
EC-s22	1	60/60; 0	≈19.9	22 / 9 / 18	25/120	n/a
ER-s22	1	60/60; 0	19.79	24 / 11 / 21	28/120	45
EE-s22	1	60/60; 0	19.79	24 / 15 / 20	23/120	46
EC-s33	1	60/60; 0	20.64	25 / 13 / 19	22/120	44
ER-s33	1	60/60; 0	19.95	23 / 13 / 17	27/120	54
EE-s33	1	60/60; 0	19.55	25 / 15 / 20	22/120	47

EC-s22's individual results ZIP was not downloaded. Its complete, valid record was read directly from Drive; its ≈ 19.9 seconds/update is approximate, and its mean output-token count is unavailable rather than imputed. ER-s33 and EE-s33 arrived in the final `Phase2_Train_logs.zip`. That archive contains the whole training root without weights and passed its checksum checks. The unchanged DEV baseline scored E/R/C $33/19/24$ out of 40, with $12/120$ strict-invalid outputs and mean output length 121 tokens.

The adapter freeze was produced on 8 November 2026 by `freeze-manifest` from the training-log archive and validated baseline archive (D051). `phase2/ADAPTERS_FROZEN`

`.md` records retained seeds 11, 22 and 33, adapter/configuration hashes, base revision and shard hashes, exposure-order identity, and training-log and baseline identities. Its SHA-256 is `1bebf10d708620b2b8f5c11aac340ca76b12e4c990dbfff9dcc82a8f9f20929f`. The implementation manifest has SHA-256 `726241c5726aaba2c51fc027866dd572abfc13d02ef9235d1901af87a9f5b43d`. The final-test gate binds the evaluated checkpoints to this freeze and rechecks the weights. All three complete paired triplets were retained, so $S = 3$ and confirmatory analysis was permitted.

9.2.2 Environment pinning and interruption recovery

On 7 November 2026, Colab's default image changed to torch 2.14.1+cu130. The notebook rejected that environment before opening an attempt, so it generated no training-attempt record. From seed 22 onward, the operator selected an older runtime image with the registered torch 2.11.0+cu130. Every retained run used the pinned stack on a T4; no code, plan or data changed. A second Google account accessed the same shared training folder for later runs; the single-writer rule applied across accounts.

The amended protocol distinguished typed infrastructure failures from numerical or protocol failures. Only the former allowed an identical from-scratch training retry; a second failure would replace the entire seed triplet using the registered reserve queue. Numerical failures, including scaler skips, and protocol failures required a halt and diagnosis. None of those training replacement rules was invoked. During evaluation, interrupted runs resumed their stored journals without regenerating successful responses. Explicit confirmation that no other session was running was required before stale-lock recovery.

Table 9.2: Recorded evaluation-lock recoveries on 10 November 2026. Times are UTC. The final column is per run, not per recovery event.

Run	Recovery time	Records kept	Final outcome
EC-s22	03:21	278	301 attempts, 300 completed; one infrastructure failure.
EE-s33	03:28; 05:41	6; 297	301 attempts, 300 completed; one infrastructure failure.

The locks followed disconnected Colab runtimes. No response was regenerated. The two affected runs each had 301 attempts for 300 completed generations; every other checkpoint had 300 attempts for 300 completed generations. Across all ten checkpoints, all 3,000 required generations resolved, with zero unresolved items. Session exports can have different ZIP/checksum hashes because derived summaries are re-exported into randomly named secondary-diagnostic folders; the journal fixes the generated responses.

Only completeness, checksums and infrastructure status were visible until all evaluations completed. The unchanged registered analysis, version `phase2_analysis 3.1.0`

(`Amendments 1--3`), ran once on 10 November 2026 (D052). It checked every run's identities, configuration, attempt budget and every run/session's T4, package and execution records. No outcome-dependent tuning or re-evaluation followed.

9.2.3 Test provenance and access disclosure

FINAL_TEST had necessarily been exposed to its drafting and verifier models and human reviewer during construction. Its provenance records single-reviewer acceptance after repairs, rather than independent human verification of every final rendering. The FT057 exception is specific: in FT057-C v4, Claude restored an omitted distractor, 12 marigolds, under delegated authority. The gold answer, 30, was checked by hand, but this revision was *not blind re-solved*. The provenance document records this rendering-level exception for the frozen `final_test.v3.json`; it must not be described as universal blind verification of the final texts.

Amendments 2–3 disclose a separate access deviation before the adapter freeze: unattended CPU tests on 3 November 2026 parsed FINAL_TEST into schema validation and scoring, embedded its bytes in synthetic notebooks, and extracted them into temporary test workspaces. These actions exceeded the byte-hash-only access rule. The affected suites were run by the Claude implementers and by Sol during review, then replaced with DEV-derived fixtures. The code shows pass/fail output and no evaluation-backbone generation. Claims that nobody viewed item content and that no information informed an analysis choice are participant attestations, not facts independently proved by the repository. Data, design and decision thresholds remained unchanged. A local commit timestamp alone is not treated as proof of precedence.

9.3 Confirmatory result

The registered strict-accuracy contrast on C was

$$\Delta_C = -4.33 \text{ pp}, \qquad 95\% \text{ CI} = [-8.00, -0.67] \text{ pp}.$$

The label is **Harm**: the unrounded interval bounds, expressed as proportions, are $L = -2/25$ and $U = -1/150$, so $U < 0$. The empirical interval is not degenerate: 24/100 families have non-zero contrasts and there are five distinct d_i values. Its bootstrap half-width is 3.67 pp. This result applies to E+C relative to E+R under the frozen recipe; it does not establish that Hinglish training is harmful under other recipes.

The Wald 95% interval is $[-8.01, -0.66]$ pp. The clipped Hoeffding 95% interval is much wider, $[-31.50, +22.83]$ pp, with radius 27.2 pp. Thus the registered approximate bootstrap label should be read alongside its conditional sampling assumptions and the

conservative bounded interval. The shared bootstrap matrix was generated with Python 3.12.7 and has SHA-256 `33f4920eab5738cf29b346be8d63a15115f9fcbda58699ee9fc44b031e46e6a1`.

Table 9.3: Registered primary bootstrap tail counts. Tails include equality and can overlap; they are not additional decision tests.

Threshold	At or below	At or above
0 pp	9921/10000	120/10000
$+5$ pp	10000/10000	0/10000

E+R's C headroom was 53.00 pp, so the observed comparator did not sit near perfect accuracy. The unchanged-DEV ceiling-risk proxy was $24/40$ (60.0%; Wilson 95% interval 44.6–73.7%), with 40.0 pp DEV headroom. It did not reach the registered $38/40$ threshold. This proxy concerns the unchanged model on DEV, not the trained comparator on FINAL_TEST.

9.4 Descriptive secondaries

All intervals in this section are descriptive. They neither add confirmatory claims nor change the strict-primary decision. Numeric-literal v1 and v2 give identical reported results here. Their C contrast is -5.67 pp $[-11.00, -0.67]$, agreeing in direction with strict accuracy.

9.4.1 S1: English retention

The five English contrasts and their strict-score flags appear in Table 9.4. Flags are independent: the within-margin criterion requires $L > -2$ pp, loss detected requires $U < 0$, and loss beyond margin requires $U < -2$ pp. They are not multiplicity-controlled non-inferiority conclusions. None meets the within-margin criterion. Every trained arm shows loss beyond the margin against the unchanged model. The registered *retention trade-off* flag additionally requires primary Support or Useful support, so it does not apply to this Harm result.

Table 9.4: English retention: estimates and 95% family-bootstrap intervals (pp). Non-zero is the strict non-zero family count. "None means no retention criterion met; "Loss; beyond means loss detected and loss beyond margin. Flags use strict only.

Contrast on E	Strict	Numeric v2	Non-zero	Flags
E+C − E+E	-2.67 $[-7.00, +1.33]$	-1.67 $[-6.33, +2.67]$	22	None
E+C − unchanged	-26.33 $[-34.33, -19.00]$	-13.67 $[-20.33, -7.00]$	37	Loss; beyond
E+R − E+E	-2.00 $[-6.33, +2.33]$	-3.67 $[-8.00, +0.67]$	26	None
E+R − unchanged	-25.67 $[-33.67, -18.00]$	-15.67 $[-23.00, -8.67]$	38	Loss; beyond
E+E − unchanged	-23.67 $[-31.67, -16.33]$	-12.00 $[-18.67, -6.00]$	38	Loss; beyond

9.4.2 S2: R transfer and S3: specificity

R transfer, T_R, compares E+R with E+E on R; Δ_R compares E+C with E+R on R. Specificity is $\Delta_C - \Delta_R$, calculated within family before averaging. The strict intervals for both R contrasts include zero. Specificity is negative, but remains a descriptive contrast, not evidence establishing a language-specific mechanism. Its bounded Hoeffding radius is 54.3 pp, with interval $[-61.99, +46.66]$ pp.

Table 9.5: R transfer and specificity: descriptive estimates and 95% family-bootstrap intervals (pp). Numeric v1 equals v2.

Contrast	Strict	Numeric v2
T_R: E+R − E+E on R	-2.33 $[-9.33, +4.33]$	-0.33 $[-7.67, +7.00]$
Δ_R: E+C − E+R on R	$+3.33$ $[-2.00, +8.67]$	$+1.67$ $[-4.67, +8.00]$
$\Delta_C - \Delta_R$	-7.67 $[-14.67, -1.33]$	-7.33 $[-14.67, -0.33]$

9.4.3 Per-seed contrasts

All three primary seed estimates are negative: -3.00, -1.00 and -9.00 pp, a range of $[-9.0, -1.0]$ pp. This sign consistency is descriptive, not a test over training seeds. English retention relative to E+E ranges from -5.0 to $+2.0$ pp, and R transfer from -6.0 to $+1.0$ pp; each has one positive and two negative seed estimates. All Δ_R seed estimates are positive, ranging from $+1.0$ to $+6.0$ pp. Table 9.6 retains the per-seed discordances and exact McNemar values without promoting them to separate confirmatory tests.

Table 9.6: Strict per-seed estimates and descriptive 95% family-bootstrap intervals (pp). First/second only count families correct only for the first/second arm in the contrast; Ret_E is E+C minus E+E on E.

Contrast	Seed	Estimate [interval]	First only	Second only	Exact p
Δ_C	11	-3.00 $[-9.00, +3.00]$	3	6	0.5078
Δ_C	22	-1.00 $[-8.00, +6.00]$	6	7	1
Δ_C	33	-9.00 $[-17.00, -2.00]$	3	12	0.03516
Ret_E	11	$+2.00$ $[-6.00, +10.00]$	10	8	0.8145
Ret_E	22	-5.00 $[-11.00, +1.00]$	2	7	0.1797
Ret_E	33	-5.00 $[-12.00, +2.00]$	4	9	0.2668
T_R	11	$+1.00$ $[-7.00, +9.00]$	9	8	1
T_R	22	-6.00 $[-15.00, +3.00]$	8	14	0.2863
T_R	33	-2.00 $[-10.00, +6.00]$	8	10	0.8145
Δ_R	11	$+1.00$ $[-5.00, +7.00]$	5	4	1
Δ_R	22	$+6.00$ $[-1.00, +13.00]$	10	4	0.1796
Δ_R	33	$+3.00$ $[-5.00, +11.00]$	10	7	0.6291

9.4.4 Accuracy by checkpoint

Tables 9.7 and 9.8 show every checkpoint's accuracy. Each cell uses 100 families and its own Wilson interval; seeds are not pooled for these intervals. Every trained checkpoint is below the unchanged model in every form under both endpoints. Strict English accuracy is 54–64% for trained checkpoints versus 85% unchanged. Numeric scoring recovers some strict errors, but the corresponding English losses against unchanged remain negative (Table 9.4); the degradation cannot be attributed wholly to strict-format scoring.

Table 9.7: Strict accuracy: correct counts, percentages and Wilson 95% intervals (%). Each interval is over families within a checkpoint.

Checkpoint	E	R	C
EE:11	62/100 (62.0%) 52.2–70.9	41/100 (41.0%) 31.9–50.8	44/100 (44.0%) 34.7–53.8
EE:22	63/100 (63.0%) 53.2–71.8	41/100 (41.0%) 31.9–50.8	41/100 (41.0%) 31.9–50.8
EE:33	59/100 (59.0%) 49.2–68.1	39/100 (39.0%) 30.0–48.8	44/100 (44.0%) 34.7–53.8
ER:11	61/100 (61.0%) 51.2–70.0	42/100 (42.0%) 32.8–51.8	44/100 (44.0%) 34.7–53.8
ER:22	56/100 (56.0%) 46.2–65.3	35/100 (35.0%) 26.4–44.7	47/100 (47.0%) 37.5–56.7
ER:33	61/100 (61.0%) 51.2–70.0	37/100 (37.0%) 28.2–46.8	50/100 (50.0%) 40.4–59.6
EC:11	64/100 (64.0%) 54.2–72.7	43/100 (43.0%) 33.7–52.8	41/100 (41.0%) 31.9–50.8
EC:22	58/100 (58.0%) 48.2–67.2	41/100 (41.0%) 31.9–50.8	46/100 (46.0%) 36.6–55.7
EC:33	54/100 (54.0%) 44.3–63.4	40/100 (40.0%) 30.9–49.8	41/100 (41.0%) 31.9–50.8
unchanged	85/100 (85.0%) 76.7–90.7	63/100 (63.0%) 53.2–71.8	66/100 (66.0%) 56.3–74.5

Table 9.8: Numeric-literal v2 accuracy: correct counts, percentages and Wilson 95% intervals (%). Each interval is over families within a checkpoint.

Checkpoint	E	R	C
EE:11	76/100 (76.0%) 66.8–83.3	52/100 (52.0%) 42.3–61.5	58/100 (58.0%) 48.2–67.2
EE:22	81/100 (81.0%) 72.2–87.5	49/100 (49.0%) 39.4–58.7	53/100 (53.0%) 43.3–62.5
EE:33	77/100 (77.0%) 67.8–84.2	54/100 (54.0%) 44.3–63.4	57/100 (57.0%) 47.2–66.3
ER:11	78/100 (78.0%) 68.9–85.0	55/100 (55.0%) 45.2–64.4	54/100 (54.0%) 44.3–63.4
ER:22	72/100 (72.0%) 62.5–79.9	48/100 (48.0%) 38.5–57.7	58/100 (58.0%) 48.2–67.2
ER:33	73/100 (73.0%) 63.6–80.7	51/100 (51.0%) 41.3–60.6	61/100 (61.0%) 51.2–70.0
EC:11	78/100 (78.0%) 68.9–85.0	55/100 (55.0%) 45.2–64.4	49/100 (49.0%) 39.4–58.7
EC:22	76/100 (76.0%) 66.8–83.3	52/100 (52.0%) 42.3–61.5	56/100 (56.0%) 46.2–65.3
EC:33	75/100 (75.0%) 65.7–82.5	52/100 (52.0%) 42.3–61.5	51/100 (51.0%) 41.3–60.6
unchanged	90/100 (90.0%) 82.6–94.5	73/100 (73.0%) 63.6–80.7	71/100 (71.0%) 61.5–79.0

9.4.5 Operation-word strata and output diagnostics

The frozen operation-word tags distinguish families in which R uses a Hindi operation word and C keeps it (*keep*), C switches it to English (*switch*), or there is no operation word (*none*). All stratum estimates of Δ_C are negative (Table 9.9). These descriptive counts do not test an interaction or isolate the effect of an operation-word substitution.

Table 9.9: Operation-word strata on C (strict). Trained-arm counts pool generations across the three retained seeds, with percentages over those generations; unchanged counts use one generation per family. These pooled counts are descriptive, not independent observations for inference.

Stratum	Families	Δ_C (pp)	E+E	E+R	E+C	Unchanged
keep	25	-1.3	43 (57.3%)	43 (57.3%)	42 (56.0%)	19 (76.0%)
none	50	-6.7	50 (33.3%)	55 (36.7%)	45 (30.0%)	32 (64.0%)
switch	25	-2.7	36 (48.0%)	43 (57.3%)	41 (54.7%)	15 (60.0%)

The execution report describes trained outputs as about 42 tokens versus about 110 for the unchanged model. For example, median C output lengths are 42.0 tokens for EC:11 and 111.0 for unchanged. Every checkpoint/form row reports zero truncated outputs. Strict-invalid reasons are led by `invalid_unit`; EC:11 on C has 24 such outputs, three `invalid_`

`answer` outputs and four `percent_requires_dimensionless` outputs. The unchanged model on C has 11 `invalid_unit`, two `invalid_answer`, one `invalid_number` and four `percent_requires_dimensionless` outputs. These are automatic scorer categories, not adjudicated explanations of mathematical errors. The source analysis preserves the full per-checkpoint/form invalid-reason, token and latency table.

9.5 Interpretation

The result does not support the proposed benefit of E+C over E+R for C accuracy under this recipe. The primary estimate is negative, its registered bootstrap interval excludes zero, and all retained seed estimates have the same negative sign. The Harm label is therefore the pre-registered reading of the observed comparison, not a revised criterion introduced after seeing the outputs.

The larger practical finding is that all trained arms degraded relative to the unchanged model. This includes E+E, so replacing English with Hinglish in the second example is not necessary for the loss. On English, strict losses against unchanged are -26.33 pp for E+C, -25.67 pp for E+R and -23.67 pp for E+E, each flagged as loss beyond margin. The unchanged model is an essential reference: comparing trained arms alone would obscure the shared deterioration.

D050 identified the target-design concern from DEV before final-test evaluation. The first EC adapter produced shorter answers (mean 43 versus 121 tokens) and more unit-related strict-invalid outputs (18 versus 8). The frozen English reference solutions were terse and shared by every training arm. The final-test pattern of short outputs, strict unit errors and lower numeric accuracy is consistent with a limitation of that shared training recipe. It does not prove that answer shortening caused the accuracy loss, nor separate target style from the other consequences of fine-tuning. No alternative target style, training schedule or controlled mediation intervention was evaluated. The defensible conclusion concerns the complete recipe; attributing the negative arm contrast to code-switching itself, or the shared loss solely to terse targets, would exceed the design.

The DEV observation did not authorize repair within the registered study. Continuing the unchanged recipe preserves the meaning of its confirmatory comparison while exposing a substantive design limitation. A future test of richer targets or another schedule would be a new study, not a correction to these scores.

9.6 Post-hoc exploratory analysis and failure audit

Evidence label: post hoc, descriptive, not pre-registered analyses. The frozen scores, inclusion rules and confirmatory label *Harm* are unchanged. The failure audit was required by Amendment 1, but the realised human check covers a subset of the failures; the final descriptive labels must not be read as human verification of every output.

9.6.1 Error decomposition

The exploratory analysis reads the ten retained checkpoint archives, checks their inner checksums, identities and paired coverage, and reproduces all 60 published strict/numeric checkpoint–form counts. It retains the stored strict scores. Table 9.10 transcribes counts and medians from `analysis/phase2_exploratory/error_analysis.json`; the accompanying README and `error_analysis.md` document the checks and interpretation limits. Trained rows pool E+E, E+R and E+C across the retained seeds. Repeated family/seed observations are descriptive counts, not independent observations for inference.

The disjoint partition is $S+F+M+U = N$: S is strict-correct; F is numeric-correct but strict-wrong; M is a numeric-literal mismatch among strict-wrong outputs; and U is numerically unscored and strict-wrong. Thus the strict deficit against unchanged decomposes into the trained-minus-unchanged rates of F, M and U, separately for each form. Numeric matching ignores units and conversions. A mismatch identifies a wrong declared literal, not its mathematical cause; an unscored answer is not evidence of a reasoning error.

Table 9.10: Post-hoc error decomposition by form. Numeric is the numeric-literal correct count; S, F, M and U form the disjoint strict-outcome partition. Tokens are median output lengths. All counts and medians are copied from the exploratory JSON.

Group	Form	N	Numeric	S	F	M	U	Tokens
Trained	E	900	686	538	171	184	7	43.0
Trained	R	900	468	359	125	402	14	42.0
Trained	C	900	497	398	114	353	35	42.0
Unchanged	E	100	90	85	8	7	0	115.0
Unchanged	R	100	73	63	11	23	3	105.5
Unchanged	C	100	71	66	8	23	3	111.0

All F outputs in both groups have strict reason `invalid_unit`: 410 trained and 27 unchanged. These are demonstrably number-correct rejected-unit answers, not necessarily semantically correct units or sound reasoning. Across trained outputs, 658 have `invalid_unit`: 410 numeric matches, 245 mismatches and three unscored answers; 487 occur where the canonical unit is empty. Rejected count nouns therefore contribute to the strict deficit, but wrong literals also remain. Numeric accuracy is not a simple superset of strict accuracy: the numeric parser rejects some suffixes accepted by the strict scorer,

leaving 54 trained and seven unchanged strict-only successes. Consequently, subtracting numeric from strict accuracy does not by itself count rejected-unit failures.

Across all forms, median output length is 42.0 tokens for trained checkpoints versus 113.0 for unchanged. The per-form medians are in Table 9.10. Shorter outputs and compressed equations are consistent with a changed response style, but this observational comparison does not establish that omitted reasoning caused the accuracy loss.

9.6.2 Failure audit of all E+R and E+C failures on C

The failure packet covers every strict-incorrect C output from every retained E+R and E+C seed: 159 E+R failures and 172 E+C failures, or 331 in total. GPT and Claude were the two AI taggers, blind to each other's labels. The agreement file records primary-category agreement on 301/331 rows, with Cohen's κ of 0.869, and format-only versus substantive agreement on 326/331, with κ of 0.96. English-status agreement was only 96/331, with near-zero κ of -0.027. The disagreement was systematic: Claude treated terse English labels attached to equations as compliant, whereas GPT generally required an English explanatory proposition and otherwise used `no_explanation`. This is an ambiguity in the minimal-English rule, not simply item-level annotation noise.

A human checker returned labels and notes for 65 rows: all 30 category disagreements, 20 sampled agreed rows and 15 sampled English-only disagreements. On category disagreements, a human checker sided with Claude on 28/30 and GPT on 1/30; the category matched both taggers on all 20 agreed rows. On the English-only sample, a human checker supported the stricter English rule on 13/15 rows. Labelled equations alone therefore count as `no_explanation` under the rule used for finalisation.

Finalisation follows human $>$ agreed $>$ `rule_adjudicated`: a checked row takes the labels from a human checker; otherwise an agreed label is retained; unchecked English-only disagreements use the stricter adjudicated rule. Categories agree on every unchecked row. The final summary records 65 rows with source `human`, 67 with `ai_agreed` and 199 with `rule_adjudicated`. This completes a descriptive classification of all failures, but the 65-row human check does not fulfil Amendment 1's requirement for human checking of every audited output.

Table 9.11: Final descriptive C-failure audit, transcribed from `results/phase2/audit/final_audit_summary.json`. Substantive is the subtotal of the four mathematical-error categories shown below format-only; subtotals are not additional failures.

Classification	E+R	E+C	Total
Format-only	46	40	86
Misread quantity	32	35	67
Wrong relation/operation	66	78	144
Arithmetic slip	7	11	18
Unit error	8	8	16
Substantive subtotal	113	132	245
All failures	159	172	331

Final English status is `no_explanation` for 232/331 failed outputs: 112 E+R and 120 E+C. The other final statuses are 95 compliant and four non-English outputs. These are failure-only counts; strict-correct outputs were not audited for English status, so they do not estimate compliance across the whole evaluation. The audit README, `results/phase2/audit/agreement.json` and `results/phase2/audit/final_audit_summary.json` preserve the annotation process, agreement and final counts.

E+R's 159 failures comprise 46 format-only and 113 substantive failures; E+C's 172 comprise 40 and 132. E+C has fewer format-only failures but more substantive failures. The E+C deficit is therefore substantive, not a formatting artifact, under this descriptive audit. These labels do not identify a causal training mechanism or replace the frozen confirmatory endpoint.

9.7 Limitations

The study is limited to Qwen3-4B, the frozen QLoRA recipe, English supervision and the registered decoding/scoring setup. The schedule was not optimized, and the shared terse targets constrain what the between-arm comparison can say about other approaches to language adaptation. There is no second backbone or pre-registered self-translation comparison.

The test set contains 100 families with one rendering per form. TRAIN was drafted by Claude and DEV/FINAL_TEST by GPT, so author/style shift accompanies the split. Single-reviewer test acceptance, sample-based TRAIN/DEV review and the FT057 exception limit claims about linguistic validity and independent verification. Shared templates and authors may violate the bootstrap's independent-family approximation. The recorded construction exposure and unattended-test access deviation also preclude describing the test as never previously accessed.

Only three seed triplets were retained. Their signs and ranges describe these check-

points; the family-bootstrap interval does not quantify uncertainty from retraining. The wide Hoeffding interval illustrates the limits of a distribution-free bounded comparison at this sample size. The planning analysis did not reliably power small gains or English retention within the narrow registered margin. Failure to meet a retention criterion against E+E is therefore not proof of equivalence or of no loss.

Strict correctness combines answer accuracy with output-format and unit requirements. Numeric-literal scoring helps describe that sensitivity, but is not a complete adjudication of reasoning quality. All retention flags, numeric endpoints, per-seed tests and operation-word strata remain descriptive. The post-hoc analysis and failure audit in Section 9.6 cannot retroactively change the confirmatory result; human checking covered only 65 of the 331 audited failures.

Sources: `phase2/PREREGISTRATION.md`, including Amendments 1–3 (methods, access and decision rules); `phase2/ADAPTERS_FROZEN.md` (retained checkpoints and identities); `phase2/FINAL_TEST_FROZEN.md` (test provenance and FT057 exception); `docs/DECISIONS.md`, D040–D052 (adoption and execution history); `results/phase2/train/README.md` (training table and DEV observations); `results/phase2/final_test/README.md` (completion and infrastructure events); and `results/phase2/final_test/phase2_analysis.md` (all final-test estimates, intervals, flags, counts and diagnostics).

Chapter 10

Limitations and Ethics

10.1 Limitations

Sample size. With 70 and 24 families, intervals are wide and the operation-word strata (17–35 families) are sparse and descriptive. Model-free power calculations showed that detecting a 5-point paired difference needs hundreds of families. The original cohort is near ceiling for Qwen, and its analysis plan was written after Pilot-B results were known.

Single-reviewer data. Pilot-B, FINAL_TEST and the lexicon were each reviewed by one bilingual reviewer, who is a team member. TRAIN and DEV received sample-based review of 45 families. The original cohort had two independent human solvers but one human language reviewer plus the adjudicator. Naturalness judgments reflect the register of very few speakers. A second-annotator sample is planned (D027).

AI-drafted items. Most renderings were drafted by an LLM, and the items may share LLM drafting styles. Claude both drafted and blind-solved Pilot-B. The GPT/Claude repair-rate contrast (68/100 vs 13/70) is confounded with split, prompt and review timing.

Review-integrity issues and how they were handled. The original pilot's review surfaced several problems. Each was handled by excluding the affected evidence, not by repairing it after the fact.

- Earlier review rounds turned out not to be independent. They were discarded and archived as unused, as were later edited copies of returned files.
- The two returned re-reviews of the five repaired families were byte-identical, so they cannot count as independent re-review. Those five families were excluded from the cohort.
- One reviewer's official stage-2 file carried an AI-prefill marker on 70 of 90 rows, a field that the review form does not write. Those judgments were excluded from all human-review and agreement claims, and the cohort's language review is described as one human plus the adjudicator (D037).
- The draft metadata labelled the author-written English statements as AI drafts. The author's correction is recorded without altering the frozen files (D038).

All exclusions were decided before any model output on the cohort, and the amended plan was committed first.

Different C definitions. The original cohort uses the clause-based C and Pilot-B the insertional C, which prevents a pooled C estimate. The original 30 drafts also have unbalanced operation words (27/30 R vs 10/30 C).

One model family per size. The three backbones are two Qwen3 sizes and one Llama model. Scale and family are not crossed, and the format screen excluded four of seven screened models whose formatting the scorer does not accept. The results are conditional on models that can follow the format.

No native-script control. There is no Devanagari arm, so an E−R gap mixes script, language, register and spelling. The pattern C $\approx$ E with R $<$ E points to Hindi vocabulary or register rather than mixing, but does not identify the cause. Each R rendering is also only one of many possible spellings.

Contamination. The items are new, but the models' pretraining data cannot be audited.

Hardware: T4 and FP16. All runs use 4-bit NF4 weights, FP16 compute, greedy decoding and no thinking mode on a free T4, which lacks bf16. The results are conditional on this configuration. Gemma-3-4B could not run under FP16 at all. No cross-hardware bitwise reproducibility is claimed.

Residual format effects and thin development evidence. Strict scoring still penalizes some correct answers (unlisted English unit phrases, the spelling *rupye*), and the secondary endpoint has the trailing-₹ gap; both are reported, not corrected. The prompt was revised after seeing outputs on 12 unreviewed fixtures. This is disclosed, and no cohort item was used to design the prompt.

Audit. The pre-registered output audit rests on two AI taggers with a human check of disagreements and a 20-row sample, not a full human audit, and it covers only the primary model on Pilot-B. The post-hoc extension to Qwen3-8B and Llama-3.2-3B used three AI taggers (Claude Opus, GPT, Claude Sonnet), and was finalised with a 97-row human check (Section 6.7). The Phase-2 failure audit likewise used two AI taggers with a 65-row human check; 199 English-status labels there were assigned by a rule fitted to that check rather than by a human (Section 9.6).

Post-hoc analyses. The tokenization analysis (Section 6.10) and the audit extension were designed after all cohort outputs had been seen. They are labelled as such, uncorrected for multiple testing and not used for any confirmatory claim.

Training study scope. The Phase-2 result holds for one backbone (Qwen3-4B), one frozen recipe (60 QLoRA updates on short English targets) and 100 FINAL_TEST families. The registered bootstrap interval is conditional on the three trained seeds and does not add seed-to-seed training variation, and the conservative Hoeffding interval is much wider ($[-31.50, +22.83]$ points). The recipe cost every arm about 24–26 points of English accuracy, so the comparison is between similarly degraded models, and a recipe with longer, explanation-rich targets might behave differently. The result says that E+C is worse than E+R under this recipe, not that Hinglish training is harmful in general.

10.2 Ethics

Consent and credit. Reviewers received a consent note covering the task, data use and release, and all reviewers consented to release of the data under CC BY 4.0. Reviewers will be credited according to their stated preferences; the adjudicator declined credit. Real identities are kept out of shared ledgers, and reviewers appear only under pseudonyms (R01, R02, A01).

Data. The items are invented everyday word problems with invented first names, and an automated scan found no contact details. Romanized Hindi varies by region and register, and these renderings reflect few speakers. FINAL_TEST was kept unreleased until its single confirmatory evaluation (10 November 2026), to protect it from contamination; an earlier automated access by unit tests, with no model outputs involved, is disclosed in pre-registration Amendment 2 (D046).

Licences. Data are released under CC BY 4.0 (`DATA_LICENSE.md`) and code under the MIT licence (`LICENSE`), with "the project authors" as copyright holder (D034). The team confirmed that DTU has no specific licence requirement.

10.3 AI-assistance disclosure

AI assistants were used extensively, and this section states how.

1. **AI-drafted items.** Renderings were drafted by an LLM: Claude for Pilot-B and TRAIN, and GPT for FINAL_TEST and DEV. For the original pilot, the English statements were written by the author, and R and C were translated by an LLM (OpenAI GPT via Codex). All families were then checked against code-computed gold and reviewed by humans, and the AI-drafted sets were also verified blind by two LLM solvers.
2. **AI-assisted human review.** Humans made every acceptance decision. Some lexicon entries and some text edits were drafted by AI before reviewers confirmed them.

In the original cohort, one reviewer's AI-prefilled, unverified stage-2 judgments were excluded.

3. **AI-assisted audit.** The output audit used two AI pre-taggers (Claude and GPT) with a human check. The post-hoc extension used Claude Opus, GPT and Claude Sonnet as taggers, with a 97-row human check. The Phase-2 failure audit used GPT and Claude Opus as taggers, blind to each other, with a 65-row human check.
4. **Writing, coding and decisions.** AI assistants helped write the evaluation and analysis code, the research notes, the paper draft and this report. Under the team's explicit delegation (D017), they also recorded protocol decisions in the decision log, where each one is labelled as an AI decision under delegation. The Phase-2 protocol draft and power simulation (D040) and the training smoke-test code (D041) were written by a Claude Opus subagent standing in for the GPT assistants, whose access had expired; the smoke-test code has no independent GPT cross-check yet. For Phase 2, Claude coordinated the runs and the analysis; GPT (Astra) drafted Chapter 9, and GPT (Sol) independently re-computed the confirmatory result from the raw predictions, with exact agreement. The team reviewed this work and takes responsibility for it.

Systems used: Claude Opus 5.5 and Claude Sonnet (Anthropic); GPT-6 Astra and GPT-6.1 Sol (OpenAI, via the Codex CLI).

Chapter 11

Conclusion and Future Work

11.1 Conclusion

With the mathematics held fixed, three small open models were least accurate on Romanized Hindi in two separately reviewed, pre-registered cohorts, by a strongly model-specific amount (about 7–9 points for Qwen, 25–37 for Llama). Both Qwen models had equal observed aggregate accuracy on English and Hinglish. The R errors were not concentrated in items with Hindi operation words, and instruction following eroded on non-English input. On excluded data, a single phrase in the answer-format instruction produced form-dependent scoring failures, and format compliance varied sharply across models. The methodological lesson is that answer formats and scorers must be checked on excluded data before any language comparison is run. A post-hoc exploratory analysis found R questions 1.4–1.6 times longer than E in tokens, but length did not account for the Qwen R deficit, so tokenization is treated as at most a minor correlate. The phase also delivered a reproducible pipeline with 161 passing tests, 300 frozen Phase-2 families, a T4 training smoke test that passed its feasibility gate, and a training schedule fixed before any training outcome.

Phase 2 then tested whether training on Hinglish helps with Hinglish. Under a frozen pre-registration, three QLoRA arms of Qwen3-4B with identical English targets were trained with three paired seeds, all nine runs valid at the first attempt, and evaluated once on 100 untouched families. E+C scored 42.7% on C against 47.0% for E+R: $\Delta_C = -4.33$ points $[-8.00, -0.67]$, the pre-registered *Harm* label, with all three seeds negative. An independent re-computation agreed exactly, and a human-checked audit found 132 substantive E+C failures on C against 113 for E+R, with slightly fewer format-only failures, so the deficit is not a scoring artifact. All trained arms fell well below the unchanged model (85/63/66% on E/R/C) and wrote much shorter outputs (median 42 vs 113 tokens). Under this recipe, Hinglish input in training did not transfer to Hinglish test accuracy, and the dominant effect of training was the loss of the model's own reasoning style.

11.2 Timeline to May 2027

Table 11.1: Status and plan for the remainder of the project (as of 10 November 2026).

Period	Milestone
Oct 2026 (done)	Original-cohort runs and analysis; SRW mentorship submission #12 (26 Oct, D039); Pilot-B audit of Qwen3-4B; post-hoc tokenization analysis and audit-extension tagging; Phase-2 protocol draft and power simulation (D040); T4 smoke test, GO (28 Oct, D042).
Nov 2026 (done to 10 Nov)	Schedule fixed (3 Nov, D043); pre-registration frozen with three amendments (D044–D047); DEV ceiling check, not fired (D049); nine training runs, all valid at the first attempt (D050–D051); adapter freeze (8 Nov); single FINAL_TEST evaluation and analysis, Harm (10 Nov, D052); human checks of both audits. Still to come: minor evaluation.
Dec 2026 – Jan 2027	SRW mentor feedback (5 Dec), final submission (15 Dec), notification (5 Jan); second-annotator sample (Pilot-B, FINAL_ TEST).
Feb – Mar 2027	Training study already complete (originally planned here). Optional new, separately pre-registered follow-up with explanation-rich targets.
Apr 2027	Thesis and training-paper draft.
May 2027	Final submission.

11.3 Publication plan

Under decision D023, the pilot findings target the EACL 2027 Student Research Workshop as a 4-page short paper. The paper was submitted to the SRW pre-submission mentorship on 26 October 2026 (OpenReview submission #12, D039); mentor feedback is due on 5 December, and the final paper needs a new submission to the SRW track by 15 December 2026. The training study targets the ARR May 2027 cycle, whose date is unconfirmed. Co-authorship with the faculty guide is to be discussed with the guide.

11.4 Further future work

Beyond the completed study, the limitations point to clear extensions:

- a separately pre-registered training recipe with explanation-rich English targets that preserves English accuracy, to test whether the Harm result is specific to the frozen

recipe;

- a reviewed Devanagari arm, to separate script from language, and independently authored spelling variants of R;
- a second independent language reviewer for every cohort;
- more model families at matched sizes;
- a full human audit of outputs;
- content-fixed interventions (lexical swaps, length padding) to test whether tokenization length plays any causal role.

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Appendix A

Frozen Prompt

The system message used for every form and every model (`test1-english-final-v2`, decision D020) is:

> `Solve the problem. Give a short explanation in English, even if the question is not in English. End with exactly one line: FINAL: <number or fraction>, followed by its unit in English only if the answer has a unit; write nothing after a pure number or fraction.`

The user message is the question text alone. Decoding is greedy and non-thinking, with a 1,024-token output cap and a fresh context per prompt.

The superseded instruction (`test1-english-final-v1`, kept in the code as `INSTRUCTION_V1`) asked for the final line in the form `FINAL: <number or fraction> <unit when applicable>`. The phrase "unit when applicable" led to unit words on dimensionless answers (Chapter 5).

Appendix B

Dataset and Run Hashes

All hashes are SHA-256. The dataset hashes were re-computed for this report and match the recorded values.

Table B.1: Frozen dataset files.

File	Families	SHA-256
`pilot_b/pilot_b_families.v2.json`	70	`e632e5fcc6b050e93d85920d7938fbb229936fa1474095d6c4c4a6517fcf3478`
`phase2/final_test.v3.json`	100	`da89dd9a626e438dffca39e3a74c4c8abc6bb57501495723b87e6ee143f3d46`
`phase2/train_A.v2.json`	80	`b9656a8c41af2a795bce9c131efc6a5efc652df78368d14fb7720fd985883666`
`phase2/train_B.v2.json`	80	`c975fe12459df6160effc5aeb0087f94666ba0b49778c3d827a67bb923ff35bd`
`phase2/dev.v2.json`	40	`e741c1652f5b3dcaebc78d9dd9d7c31fdf4417306fbb2f0862149f27f858f28b`
`pilot_original/cohort_dataset.json`	24	`5c3e8746f0629d1908902658bda2c280aeb388c55e741fb8e612bc49b5d43103`
`pilot_original/cohort_splits.json`	–	`8da8f40eaa01932152032c290f32e20345b7b22092104c0c329d63d0df0067cc`

The original cohort's canonical-JSON engine digests are `02752d35b6bde18392afb2e02fa3cdc430437801606706a35662bc62fe2ad03a` (dataset) and `8f5996187c291bae21d75fbccb83486ce18bad3b9a13feaaad9fda20a2628433` (splits). Its source draft, `pilot/annotation_draft_v2/coordinator/pilot_dataset.DRAFT.json`, has hash `9a4434b67941505e9a3679ba6cf0c344f50804230e4fd3591df1f3e192d4711a`.

Development fixture runs (FMT2) have returned ZIP hashes recorded in their `research/fmt2_*/README.md` files; for example, the Qwen3-4B prompt-v1 run is `c21741273a1f1c825bc141957a30439b95d567f456e9fc68da0f83e0a004e84f` and the prompt-v2 run is `28794810b54579a6b78f61b36b9354d02eb338cf481dede5869791e7c5cb36d6`.

Table B.2: Returned cohort results ZIPs and pinned model revisions.

Cohort	Model	Returned ZIP SHA-256
Pilot-B	Qwen3-4B	`7d9274d4af0d4c7599cb68f8353ee6cd695a5d8acd2c22a5facd738f7076e7c5`
Pilot-B	Qwen3-8B	`3d1cef58a01116caf7a81f8ace5b641096034735b3e9655267f501ac62455180`
Pilot-B	Llama-3.2-3B	`08120040e841b1208f708a83ceec145c02316e2352d1bc8b7a0a29ae39792458`
Original	Qwen3-4B	`ff2af2bfd51d3ae301e2c6549f390c8953501c199fe5255ed7a76a19865cab37`
Original	Qwen3-8B	`403d7280558d8ba26553d0e07f638650ad87edc08c8b34454f1d0895d1e4509c`
Original	Llama-3.2-3B	`5d9b5f657f6cbb147b7adf1b1b84f60414e7fde018d2accb03be21379dd99f31`

Model	Pinned revision
`Qwen/Qwen3-4B`	`1cfa9a7208912126459214e8b04321603b3df60c`
`Qwen/Qwen3-8B`	`b968826d9c46dd6066d109eabc6255188de91218`
`meta-llama/Llama-3.2-3B-Instruct`	`0cb88a4f764b7a12671c53f0838cd831a0843b95`

Appendix C

Decision Log Summary

The full log is `docs/DECISIONS.md`. It distinguishes user instructions, human decisions, AI recommendations and AI decisions taken under the student's delegation (D017).

ID	Summary
D001	Essential eventual arms E+E/E+R/E+C, matching content, exposure and full English targets.
D002	Pilot design: unchanged model, 30 families/90 renderings; human acceptance required.
D003–D008	Early process: annotation budget open; review packets distributed; shared-folder coordination; frozen scorer v1 and packets kept unchanged; reviewers not yet done.
D009–D011	Secondary-endpoint specification proposed; currency-prefix and unit-alias handling left to human decision.
D012–D014	One writer at a time; each work cycle must change data/code/results or end with a concrete question; excluded format-test suite proposed.
D015	Training margins (+5pp gain, -2pp English), three seeds, 256 updates: discussion proposals, NOT APPROVED (update count superseded by D043).
D016–D017	Minor evaluation November 2026, final May 2027; the team delegated spec-adoption and format-test decisions to Claude; goal is a publishable paper.
D018	Adopted `numeric_literal_v1` as secondary endpoint beside strict v1.0.0; audit taxonomy; CMI_v1; consent note.
D019–D020	FMT2 format run on excluded fixtures; prompt replaced by `test1-english-final-v2` before any reviewed generation.
D021	Training-stack plan and go/no-go gates (56 s / 150 s per update).
D022	CSV formula-guard bug fixed before any pilot run.
D023	Publication plan: EACL 2027 SRW (pilot), ARR May 2027 (training).
D024–D026	Phi-4-mini tested and excluded (inline FINAL, R repetition); Granite excluded (commentary after FINAL).
D027	Scale-up: 70-family Pilot-B and 300 Phase-2 families; AI drafting, code gold, Claude/GPT blind solving, human review.
D028	Screen four more models; pre-declared inclusion rule $\leq$3/36 strict-invalid.

ID	Summary
D029–D032	Qwen3-1.7B excluded; Qwen3-8B included; Gemma-3-4B excluded; Llama-3.2-3B included.
D033	Pilot-B review returned; C redefined as insertional for Pilot-B and Phase 2; R borrowing list extended.
D034	Licences: code MIT, data CC BY 4.0.
D035	Pilot-B evaluated in `pilot_b` mode under a plan committed before outputs.
D036	Original cohort = 24 families accepted without repair; excluded PIL010 and five repaired families.
D037	R01's AI-prefilled stage-2 judgments excluded; cohort relabelled (two human solvers; language review R02 + A01); plan amended before outputs.
D038	Provenance correction: original E statements author-written; R/C AI-translated, then human-reviewed.
D039	Pilot paper submitted to the EACL 2027 SRW pre-submission mentorship (#12); final SRW submission due 15 Dec 2026.
D040	Phase-2 protocol direction adopted for the pre-registration draft: Δ_C as an estimate with a precision target; decision labels; -2pp retention margin; `numeric_literal_v2` as new secondary; DEV ceiling flag; self-translate control; no Devanagari arm.
D041	Training smoke-test code and notebook built; 16 TRAIN families fixed for the smoke test; adapters never evaluated.
D042	Smoke test passed on a Colab T4 (worst-arm median 20.6 s/update, exact reload); training stays on Colab with Qwen3-4B; memorisation within about 12 passes flags 256 updates as over-fitting risk.
D043	Training schedule fixed: 160 TRAIN families, 60 updates (3 passes), lr 10^{-4} with warm-up and cosine decay, seeds 11/22/33, checkpoint after update 60.
D044–D047	Phase-2 pre-registration frozen (3 Nov) and amended three times before any Phase-2 output, after GPT adversarial reviews; Amendment 2 discloses an automated FINAL_TEST access by unit tests.
D048	Implementation freeze after an end-to-end CPU rehearsal.
D049	Unchanged-model DEV baseline 33/19/24 of 40; ceiling flag not fired.
D050	First trained adapter valid; shorter outputs and more unit-related strict failures on DEV noted.
D051	All nine runs valid at attempt 1; S = 3, confirmatory; adapters frozen (8 Nov).

ID	Summary
D052	FINAL_TEST complete; $\Delta_C = -4.33$ pp $[-8.00, -0.67]$, Harm (10 Nov).

Appendix D

The Review Forms

The original pilot's review used self-contained HTML forms distributed in per-reviewer ZIP packets; reviewers returned the JSON files the forms exported.

[*Optional: insert a screenshot of the stage-1 and stage-2 forms here, taken from a copy of the packet with artificial answers (never a real reviewer's file).*]

Stage 1 (blind solving). The form presents all 90 renderings in randomized order without any reference answer. For each rendering the reviewer types a numeric answer, and the form records a timestamp. Progress can be downloaded as JSON and resumed by import. When all 90 are solved, the reviewer completes an attestation and downloads `R01_stage1_responses.json` (or R02).

Stage 2 (language review). This form refuses to save anything before a stage-1 file is imported, and it rejects an unfinished stage-1 file. On import it keeps the stage-1 answers read-only and records the stage-1 file's SHA-256. For each E/R/C triplet the reviewer then rates equivalence, ambiguity, classification (whether R and C meet their definitions) and naturalness, and chooses a disposition. Each reviewer's stage-2 form accepts only that reviewer's own stage-1 file. The form does not write any AI-provenance field.

Adjudication. The adjudicator's packet, with the reference answers and the decision worksheet, is released only after both independent submissions are collected. The adjudicator records ACCEPT, REVISE or EXCLUDE per family, together with gold and unit decisions. Repaired texts require new versions and an independent re-review.

Pilot-B and Phase 2. These sets were reviewed through the `autoverify` review sheet (a CSV with E/R/C texts, flags, `equivalence_ok`, naturalness ratings, a decision and notes), with the gold column placed last so it can be hidden until the reviewer has assessed the question.

Appendix E

Reproducibility Instructions

Re-running a cohort evaluation. Each notebook in `colab_notebooks/pilot_b_eval/` and `colab_notebooks/pilot_original_eval/` is self-contained and embeds its frozen data.

1. Open `https://colab.research.google.com`, choose File → Upload notebook, and select, for example, `PilotB_Qwen3-4B.ipynb`.
2. Choose Runtime → Change runtime type → T4 GPU.
3. For Llama, add a Hugging Face token as a Colab secret, since the model is gated.
4. Choose Runtime → Run all. A results ZIP downloads at the end. If the session disconnects, rerun with the same run name: the journal resumes and skips completed generations.

Greedy decoding with pinned revisions should reproduce the outputs on the same stack, but no cross-hardware bitwise guarantee is made. A re-run must not overwrite the original returned ZIPs.

Rebuilding the tables. From the repository root, using only the Python standard library (matplotlib optional, for figures):

```
python3 -B -m paper_analysis.build_tables RUN.zip --out NEW_DIR
python3 -B results/pilot_b/strata.py \
    results/pilot_b/qwen3-4b/per_rendering.csv
python3 -B report/figures/make_figures.py
```

`build_tables` verifies every checksum and exits with code 3 if any recomputed value disagrees with the stored metrics. `--out` must be a new or empty directory.

Re-running the software tests.

```
python3 -B -m unittest discover -s pilot/scripts -p 'test_*.py'
python3 -B -m unittest discover -s t4_smoke -p 'test_*.py'
PYTHONPATH=test1_pilot python3 -B -m unittest discover \
    -s test1_pilot/tests -p 'test_*.py'
python3 -B -m unittest discover -s secondary_analysis/tests
python3 -B -m unittest discover -s paper_analysis/tests
python3 -B -m unittest discover -s autoverify/tests
python3 -B -m unittest discover -s authoring/tests
```

Blind verification of new families.

```
python3 -B authoring/check_family.py my_families.json
python3 -B autoverify/make_solver_tasks.py families.json \
    --out-dir /tmp/blind-tasks --seed 20261015
python3 -B autoverify/compare.py families.json \
    --claude claude.jsonl --gpt gpt.jsonl --out-dir /tmp/verification
```

Verifying the frozen data. `shasum -a 256` on each file in Table B.1 must reproduce the listed hash.